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Enhanced SSB beam reporting

Abstract

Systems, apparatuses, methods, and computer programs for reference signal measurement reporting by a user equipment (UE) in a wireless communications system. In one aspect, the method can include actions of measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams from a base station, determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold, responsive to the determination, calculating, based at least in part on the measured RSRP values and a horizontal/vertical (H/V) polarization imbalance for each of the plurality of SSB beams, a ranking of the plurality of SSB beams, and reporting at least one of the ranked plurality of SSB beams to the base station.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/247,788 filed Sep. 23, 2021, which is herein incorporated by reference in its entirety.

BACKGROUND

(1) Wireless communication networks provide integrated communication platforms and telecommunication services to wireless user devices. Example telecommunication services include telephony, data (e.g., voice, audio, and/or video data), messaging, internet-access, and/or other services. The wireless communication networks have wireless access nodes that exchange wireless signals with the wireless user devices using wireless network protocols, such as protocols described in various telecommunication standards promulgated by the Third Generation Partnership Project (3GPP). Example wireless communication networks include code division multiple access (CDMA) networks, time division multiple access (TDMA) networks, frequency-division multiple access (FDMA) networks, orthogonal frequency-division multiple access (OFDMA) networks, Long Term Evolution (LTE), and Fifth Generation New Radio (5G NR). The wireless communication networks facilitate mobile broadband service using technologies such as OFDM, multiple input multiple output (MIMO), advanced channel coding, massive MIMO, beamforming, and/or other features.

SUMMARY

(2) The present disclosure is directed towards systems, apparatuses, methods, and computer programs for enhanced SSB Beam RSRP reporting based on a horizontal/vertical (H/V) polarization imbalance associated with one or more beams.

(3) In particular, the present disclosure is directed to user equipment that is configured to determine that a highest RSRP value for set of reference signals associated with a plurality of beams from a base station fails to satisfy a predetermined threshold. Then, based on that determination, the UE calculates a horizontal/vertical (H/V) polarization imbalance for each of the plurality of beams. The UE then ranks each of the plurality of beams based on the calculated H/V polarization imbalance. The UE then reports, to a base station, one or more of the ranked beams. In some implementations, based on the received reporting, the base station can then switch the UE to a different beam based on the received reporting in order to sustain NR performance. The different beam can be a better beam than the UE was previously using, with a better beam being a beam that has, e.g., a higher RSRP value.

(4) In accordance with one aspect of the present disclosure, a method for reference signal measurement reporting by a user equipment (UE) in a wireless communications system is disclosed. In one aspect, the method can include measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams from a base station, determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold, responsive to the determination, calculating, based at least in part on the measured RSRP values and a horizontal/vertical (H/V) polarization imbalance for each of the plurality of SSB beams, a ranking of the plurality of SSB beams, and reporting at least one of the ranked plurality of SSB beams to the base station.

(5) Other aspects includes apparatuses, systems, and computer programs for performing the actions of the aforementioned method.

(6) The details of one or more embodiments of these systems and methods are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of these systems and methods will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a wireless communication system.
- (2) FIG. 2 illustrates an example of data obtained via a lab test indicating that a UE switched to a beam having high H/V polarization imbalance, leading to poor DL performance and link failure.
- (3) FIG. 3 illustrates an example of data obtained via a field test for pedestrian mobility indicating that a UE switched to a beam having high H/V polarization imbalance, leading to poor DL performance and link failure.
- (4) FIG. 4 is a flowchart of an example of a process for enhanced SSB Beam RSRP reporting according to one aspect of the present disclosure.
- (5) FIG. 5 is a flowchart of another example of a process for enhanced SSB Beam RSRP reporting according to one aspect of the present disclosure.
- (6) FIG. 6 illustrates an example architecture of a system including a first CN, in accordance with various embodiments.
- (7) FIG. 7 illustrates an architecture of a system including a second CN in accordance with various embodiments.
- (8) FIG. 8 illustrates an example of infrastructure equipment in accordance with various embodiments.
- (9) FIG. 9 illustrates an example of a platform or device in accordance with various embodiments.
- (10) FIG. 10 illustrates example components of baseband circuitry and radio front end modules (RFEM) in accordance with various embodiments.
- (11) FIG. 11 illustrates various protocol functions that may be implemented in a wireless communication device in accordance with various embodiments.
- (12) FIG. 12 illustrates components of a core network in accordance with various embodiments.
- (13) FIG. 13 is a block diagram illustrating components, according to some example embodiments, of a system to support NFV.
- (14) FIG. 14 is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

DETAILED DESCRIPTION

- (15) The present disclosure is directed towards systems, apparatuses, methods, and computer programs for enhanced SSB Beam RSRP reporting based on a horizontal/vertical (H/V) polarization imbalance associated with one or more beams.
- (16) In particular, the present disclosure is directed to user equipment that is configured to determine that a highest RSRP value for set of reference signals associated with a plurality of beams from a base station fails to satisfy a predetermined threshold. Then, based on that determination, the UE calculates a horizontal/vertical (H/V) polarization imbalance for each of the plurality of beams. The UE then ranks each of the plurality of beams based on the calculated H/V polarization imbalance. The UE then reports, to a base station, one or more of the ranked beams. In some implementations, based on the received reporting, the base station can then switch the UE to a different beam based on the received reporting in order to sustain NR performance. The different beam can be a better beam than the UE was previously using, with a better beam being a beam that has, e.g., a higher RSRP value.
- (17) FIG. 1 illustrates an example of a wireless communication system **100**. For purposes of convenience and without limitation, the example system **100** is described in the context of Long Term Evolution (LTE) and Fifth Generation (5G) New Radio (NR) communication standards as defined by the Third Generation Partnership Project (3GPP) technical specifications. More

specifically, the wireless communication system **100** is described in the context of a Non-Standalone (NSA) networks that incorporate both LTE and NR, for example, E-UTRA (Evolved Universal Terrestrial Radio Access)-NR Dual Connectivity (EN-DC) networks, and NE-DC networks. However, the wireless communication system **100** may also be a Standalone (SA) network that incorporates only NR. Furthermore, other types of communication standards are possible, including future 3GPP systems (e.g., Sixth Generation (6G)) systems, IEEE 802.16 protocols (e.g., WMAN, WiMAX, etc.), or the like.

(18) As shown by FIG. 1, the system **100** includes UE **101a** and UE **101b** (collectively referred to as “UEs **101**” or “UE **101**”). In this example, UEs **101** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device, such as consumer electronics devices, cellular phones, smartphones, feature phones, tablet computers, wearable computer devices, personal digital assistants (PDAs), pagers, wireless handsets, desktop computers, laptop computers, in-vehicle infotainment (IVI), in-car entertainment (ICE) devices, an Instrument Cluster (IC), head-up display (HUD) devices, onboard diagnostic (OBD) devices, dashtop mobile equipment (DME), mobile data terminals (MDTs), Electronic Engine Management System (EEMS), electronic/engine control units (ECUs), electronic/engine control modules (ECMs), embedded systems, microcontrollers, control modules, engine management systems (EMS), networked or “smart” appliances, MTC devices, M2M, IoT devices, and/or the like.

(19) In some embodiments, any of the UEs **101** may be IoT UEs, which may comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. An IoT UE can utilize technologies such as M2M or MTC for exchanging data with an MTC server or device via a PLMN, ProSe or D2D communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network describes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

(20) The UEs **101** may be configured to connect, for example, communicatively couple, with RAN **110**. In embodiments, the RAN **110** may be an NG RAN or a 5G RAN, an E-UTRAN, or a legacy RAN, such as a UTRAN or GERAN. As used herein, the term “NG RAN” or the like may refer to a RAN **110** that operates in an NR or 5G system **100**, and the term “E-UTRAN” or the like may refer to a RAN **110** that operates in an LTE or 4G system **100**. The UEs **101** utilize connections (or channels) **103** and **104**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below).

(21) In this example, the connections **103** and **104** are illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols, such as a GSM protocol, a CDMA network protocol, a PTT protocol, a POC protocol, a UMTS protocol, a 3GPP LTE protocol, an Advanced long term evolution (LTE-A) protocol, a LTE-based access to unlicensed spectrum (LTE-U), a 5G protocol, a NR protocol, an NR-based access to unlicensed spectrum (NR-U) protocol, and/or any of the other communications protocols discussed herein. In embodiments, the UEs **101** may directly exchange communication data via a ProSe interface **105**. The ProSe interface **105** may alternatively be referred to as a SL interface **105** and may comprise one or more logical channels, including but not limited to a PSCCH, a PSSCH, a PSDCH, and a PSBCH.

(22) The UE **101b** is shown to be configured to access an AP **106** (also referred to as “WLAN node **106**,” “WLAN **106**,” “WLAN Termination **106**,” “WT **106**” or the like) via connection **107**. The connection **107** can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein the AP **106** would comprise a wireless fidelity (Wi-Fi®) router. In this example, the AP **106** is shown to be connected to the Internet without connecting to the core

network of the wireless system (described in further detail below). In various embodiments, the UE **101b**, RAN **110**, and AP **106** may be configured to utilize LWA operation and/or LWIP operation. The LWA operation may involve the UE **101b** in RRC_CONNECTED being configured by a RAN node **111a-b** to utilize radio resources of LTE and WLAN. LWIP operation may involve the UE **101b** using WLAN radio resources (e.g., connection **107**) via IPsec protocol tunneling to authenticate and encrypt packets (e.g., IP packets) sent over the connection **107**. IPsec tunneling may include encapsulating the entirety of original IP packets and adding a new packet header, thereby protecting the original header of the IP packets.

(23) The RAN **110** can include one or more AN nodes or RAN nodes **111a** and **111b** (collectively referred to as “RAN nodes **111**” or “RAN node **111**”) that enable the connections **103** and **104**. As used herein, the terms “access node,” “access point,” or the like may describe equipment that provides the radio baseband functions for data and/or voice connectivity between a network and one or more users. These access nodes can be referred to as BS, gNBs, RAN nodes, eNBs, NodeBs, RSUs, TRxPs or TRPs, and so forth, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). As used herein, the term “NG RAN node” or the like may refer to a RAN node **111** that operates in an NR or 5G system **100** (for example, a gNB), and the term “E-UTRAN node” or the like may refer to a RAN node **111** that operates in an LTE or 4G system **100** (e.g., an eNB). According to various embodiments, the RAN nodes **111** may be implemented as one or more of a dedicated physical device such as a macrocell base station, and/or a low power (LP) base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

(24) In some embodiments, all or parts of the RAN nodes **111** may be implemented as one or more software entities running on server computers as part of a virtual network, which may be referred to as a CRAN and/or a virtual baseband unit pool (vBBUP). In these embodiments, the CRAN or vBBUP may implement a RAN function split, such as a PDCP split wherein RRC and PDCP layers are operated by the CRAN/vBBUP and other L2 protocol entities are operated by individual RAN nodes **111**; a MAC/PHY split wherein RRC, PDCP, RLC, and MAC layers are operated by the CRAN/vBBUP and the PHY layer is operated by individual RAN nodes **111**; or a “lower PHY” split wherein RRC, PDCP, RLC, MAC layers and upper portions of the PHY layer are operated by the CRAN/vBBUP and lower portions of the PHY layer are operated by individual RAN nodes **111**. This virtualized framework allows the freed-up processor cores of the RAN nodes **111** to perform other virtualized applications. In some implementations, an individual RAN node **111** may represent individual gNB-DUs that are connected to a gNB-CU via individual F1 interfaces (not shown by FIG. 1). In these implementations, the gNB-DUs may include one or more remote radio heads or RFEMs (see, e.g., FIG. 8), and the gNB-CU may be operated by a server that is located in the RAN **110** (not shown) or by a server pool in a similar manner as the CRAN/vBBUP. Additionally or alternatively, one or more of the RAN nodes **111** may be next generation eNBs (ng-eNBs), which are RAN nodes that provide E-UTRA user plane and control plane protocol terminations toward the UEs **101**, and are connected to a 5GC (e.g., CN **720** of FIG. 7) via an NG interface (discussed infra).

(25) In V2X scenarios one or more of the RAN nodes **111** may be or act as RSUs. The term “Road Side Unit” or “RSU” may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable RAN node or a stationary (or relatively stationary) UE, where an RSU implemented in or by a UE may be referred to as a “UE-type RSU,” an RSU implemented in or by an eNB may be referred to as an “eNB-type RSU,” an RSU implemented in or by a gNB may be referred to as a “gNB-type RSU,” and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs **101** (vUEs **101**). The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics,

media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may operate on the 5.9 GHz Direct Short Range Communications (DSRC) band to provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may operate on the cellular V2X band to provide the aforementioned low latency communications, as well as other cellular communications services. Additionally or alternatively, the RSU may operate as a Wi-Fi hotspot (2.4 GHz band) and/or provide connectivity to one or more cellular networks to provide uplink and downlink communications. The computing device(s) and some or all of the radiofrequency circuitry of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller and/or a backhaul network.

(26) Any of the RAN nodes **111** can terminate the air interface protocol and can be the first point of contact for the UEs **101**. In some embodiments, any of the RAN nodes **111** can fulfill various logical functions for the RAN **110** including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management and data packet scheduling, and mobility management.

(27) In embodiments, the UEs **101** can be configured to communicate using OFDM communication signals with each other or with any of the RAN nodes **111** over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an OFDMA communication technique (e.g., for downlink communications) or a SC-FDMA communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the embodiments is not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

(28) In some embodiments, a downlink resource grid can be used for downlink transmissions from any of the RAN nodes **111** to the UEs **101**, while uplink transmissions can utilize similar techniques. The grid can be a time-frequency grid, called a resource grid or time-frequency resource grid, which is the physical resource in the downlink in each slot. Such a time-frequency plane representation is a common practice for OFDM systems, which makes it intuitive for radio resource allocation. Each column and each row of the resource grid corresponds to one OFDM symbol and one OFDM subcarrier, respectively. The duration of the resource grid in the time domain corresponds to one slot in a radio frame. The smallest time-frequency unit in a resource grid is denoted as a resource element. Each resource grid comprises a number of resource blocks, which describe the mapping of certain physical channels to resource elements. Each resource block comprises a collection of resource elements; in the frequency domain, this may represent the smallest quantity of resources that currently can be allocated. There are several different physical downlink channels that are conveyed using such resource blocks.

(29) According to various embodiments, the UEs **101** and the RAN nodes **111** communicate data (for example, transmit and receive) data over a licensed medium (also referred to as the “licensed spectrum” and/or the “licensed band”) and an unlicensed shared medium (also referred to as the “unlicensed spectrum” and/or the “unlicensed band”). The licensed spectrum may include channels that operate in the frequency range of approximately 400 MHz to approximately 3.8 GHz, whereas the unlicensed spectrum may include the 5 GHz band. NR in the unlicensed spectrum may be referred to as NR-U, and LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

(30) To operate in the unlicensed spectrum, the UEs **101** and the RAN nodes **111** may operate using LAA, eLAA, and/or feLAA mechanisms. In these implementations, the UEs **101** and the RAN nodes **111** may perform one or more known medium-sensing operations and/or carrier-sensing operations in order to determine whether one or more channels in the unlicensed spectrum is unavailable or otherwise occupied prior to transmitting in the unlicensed spectrum. The medium/carrier sensing operations may be performed according to a listen-before-talk (LBT)

protocol.

(31) LBT is a mechanism whereby equipment (for example, UEs **101** RAN nodes **111**, etc.) senses a medium (for example, a channel or carrier frequency) and transmits when the medium is sensed to be idle (or when a specific channel in the medium is sensed to be unoccupied). The medium sensing operation may include CCA, which utilizes at least ED to determine the presence or absence of other signals on a channel in order to determine if a channel is occupied or clear. This LBT mechanism allows cellular/LAA networks to coexist with incumbent systems in the unlicensed spectrum and with other LAA networks. ED may include sensing RF energy across an intended transmission band for a period of time and comparing the sensed RF energy to a predefined or configured threshold.

(32) Typically, the incumbent systems in the 5 GHz band are WLANs based on IEEE 802.11 technologies. WLAN employs a contention-based channel access mechanism, called CSMA/CA. Here, when a WLAN node (e.g., a mobile station (MS) such as UE **101**, AP **106**, or the like) intends to transmit, the WLAN node may first perform CCA before transmission. Additionally, a backoff mechanism is used to avoid collisions in situations where more than one WLAN node senses the channel as idle and transmits at the same time. The backoff mechanism may be a counter that is drawn randomly within the CWS, which is increased exponentially upon the occurrence of collision and reset to a minimum value when the transmission succeeds. The LBT mechanism designed for LAA is somewhat similar to the CSMA/CA of WLAN. In some implementations, the LBT procedure for DL or UL transmission bursts including PDSCH or PUSCH transmissions, respectively, may have an LAA contention window that is variable in length between X and Y ECCA slots, where X and Y are minimum and maximum values for the CWSs for LAA. In one example, the minimum CWS for an LAA transmission may be 9 microseconds (s); however, the size of the CWS and a MCOT (for example, a transmission burst) may be based on governmental regulatory requirements.

(33) The LAA mechanisms are built upon CA technologies of LTE-Advanced systems. In CA, each aggregated carrier is referred to as a CC. A CC may have a bandwidth of 1.4, 3, 5, 10, 15 or 20 MHz and a maximum of five CCs can be aggregated, and therefore, a maximum aggregated bandwidth is 100 MHz. In FDD systems, the number of aggregated carriers can be different for DL and UL, where the number of UL CCs is equal to or lower than the number of DL component carriers. In some cases, individual CCs can have a different bandwidth than other CCs. In TDD systems, the number of CCs as well as the bandwidths of each CC is usually the same for DL and UL.

(34) CA also comprises individual serving cells to provide individual CCs. The coverage of the serving cells may differ, for example, because CCs on different frequency bands will experience different pathloss. A primary service cell or PCell may provide a PCC for both UL and DL, and may handle RRC and NAS related activities. The other serving cells are referred to as SCells, and each SCell may provide an individual SCC for both UL and DL. The SCCs may be added and removed as required, while changing the PCC may require the UE **101** to undergo a handover. In LAA, eLAA, and feLAA, some or all of the SCells may operate in the unlicensed spectrum (referred to as “LAA SCells”), and the LAA SCells are assisted by a PCell operating in the licensed spectrum. When a UE is configured with more than one LAA SCell, the UE may receive UL grants on the configured LAA SCells indicating different PUSCH starting positions within a same subframe.

(35) The PDSCH carries user data and higher-layer signaling to the UEs **101**. The PDCCH carries information about the transport format and resource allocations related to the PDSCH channel, among other things. It may also inform the UEs **101** about the transport format, resource allocation, and HARQ information related to the uplink shared channel. Typically, downlink scheduling (assigning control and shared channel resource blocks to the UE **101b** within a cell) may be performed at any of the RAN nodes **111** based on channel quality information fed back from any of

the UEs **101**. The downlink resource assignment information may be sent on the PDCCH used for (e.g., assigned to) each of the UEs **101**.

(36) The PDCCH uses CCEs to convey the control information. Before being mapped to resource elements, the PDCCH complex-valued symbols may first be organized into quadruplets, which may then be permuted using a sub-block interleaver for rate matching. Each PDCCH may be transmitted using one or more of these CCEs, where each CCE may correspond to nine sets of four physical resource elements known as REGs. Four Quadrature Phase Shift Keying (QPSK) symbols may be mapped to each REG. The PDCCH can be transmitted using one or more CCEs, depending on the size of the DCI and the channel condition. There can be four or more different PDCCH formats defined in LTE with different numbers of CCEs (e.g., aggregation level, $L=1, 2, 4$, or 8).

(37) Some embodiments may use concepts for resource allocation for control channel information that are an extension of the above-described concepts. For example, some embodiments may utilize an EPDCCH that uses PDSCH resources for control information transmission. The EPDCCH may be transmitted using one or more ECCEs. Similar to above, each ECCE may correspond to nine sets of four physical resource elements known as an EREGs. An ECCE may have other numbers of EREGs in some situations.

(38) The RAN nodes **111** may be configured to communicate with one another via interface **112**. In embodiments where the system **100** is an LTE system (e.g., when CN **120** is an EPC **620** as in FIG. **6**), the interface **112** may be an X2 interface **112**. The X2 interface may be defined between two or more RAN nodes **111** (e.g., two or more eNBs and the like) that connect to EPC **120**, and/or between two eNBs connecting to EPC **120**. In some implementations, the X2 interface may include an X2 user plane interface (X2-U) and an X2 control plane interface (X2-C). The X2-U may provide flow control mechanisms for user data packets transferred over the X2 interface, and may be used to communicate information about the delivery of user data between eNBs. For example, the X2-U may provide specific sequence number information for user data transferred from a MeNB to an SeNB; information about successful in sequence delivery of PDCP PDUs to a UE **101** from an SeNB for user data; information of PDCP PDUs that were not delivered to a UE **101**; information about a current minimum desired buffer size at the SeNB for transmitting to the UE user data; and the like. The X2-C may provide intra-LTE access mobility functionality, including context transfers from source to target eNBs, user plane transport control, etc.; load management functionality; as well as inter-cell interference coordination functionality.

(39) In embodiments where the system **100** is a 5G or NR system (e.g., when CN **120** is a 5GC **720** as in FIG. **7**), the interface **112** may be an Xn interface **112**. The Xn interface is defined between two or more RAN nodes **111** (e.g., two or more gNBs and the like) that connect to 5GC **120**, between a RAN node **111** (e.g., a gNB) connecting to 5GC **120** and an eNB, and/or between two eNBs connecting to 5GC **120**. In some implementations, the Xn interface may include an Xn user plane (Xn-U) interface and an Xn control plane (Xn-C) interface. The Xn-U may provide non-guaranteed delivery of user plane PDUs and support/provide data forwarding and flow control functionality. The Xn-C may provide management and error handling functionality, functionality to manage the Xn-C interface; mobility support for UE **101** in a connected mode (e.g., CM-CONNECTED) including functionality to manage the UE mobility for connected mode between one or more RAN nodes **111**. The mobility support may include context transfer from an old (source) serving RAN node **111** to new (target) serving RAN node **111**; and control of user plane tunnels between old (source) serving RAN node **111** to new (target) serving RAN node **111**. A protocol stack of the Xn-U may include a transport network layer built on Internet Protocol (IP) transport layer, and a GTP-U layer on top of a UDP and/or IP layer(s) to carry user plane PDUs. The Xn-C protocol stack may include an application layer signaling protocol (referred to as Xn Application Protocol (Xn-AP)) and a transport network layer that is built on SCTP. The SCTP may be on top of an IP layer, and may provide the guaranteed delivery of application layer messages. In the transport IP layer, point-to-point transmission is used to deliver the signaling PDUs. In other

implementations, the Xn-U protocol stack and/or the Xn-C protocol stack may be same or similar to the user plane and/or control plane protocol stack(s) shown and described herein.

(40) The RAN **110** is shown to be communicatively coupled to a core network-in this embodiment, core network (CN) **120**. The CN **120** may comprise a plurality of network elements **122**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UEs **101**) who are connected to the CN **120** via the RAN **110**. The components of the CN **120** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some embodiments, NFV may be utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **120** may be referred to as a network slice, and a logical instantiation of a portion of the CN **120** may be referred to as a network sub-slice. NFV architectures and infrastructures may be used to virtualize one or more network functions, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

(41) Generally, the application server **130** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS PS domain, LTE PS data services, etc.). The application server **130** can also be configured to support one or more communication services (e.g., VoIP sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs **101** via the EPC **120**.

(42) In embodiments, the CN **120** may be a 5GC (referred to as “5GC **120**” or the like), and the RAN **110** may be connected with the CN **120** via an NG interface **113**. In embodiments, the NG interface **113** may be split into two parts, an NG user plane (NG-U) interface **114**, which carries traffic data between the RAN nodes **111** and a UPF, and the S1 control plane (NG-C) interface **115**, which is a signaling interface between the RAN nodes **111** and AMFs. Embodiments where the CN **120** is a 5GC **120** are discussed in more detail with regard to FIG. 7.

(43) In embodiments, the CN **120** may be a 5G CN (referred to as “5GC **120**” or the like), while in other embodiments, the CN **120** may be an EPC). Where CN **120** is an EPC (referred to as “EPC **120**” or the like), the RAN **110** may be connected with the CN **120** via an S1 interface **113**. In embodiments, the S1 interface **113** may be split into two parts, an S1 user plane (S1-U) interface **114**, which carries traffic data between the RAN nodes **111** and the S-GW, and the S1-MME interface **115**, which is a signaling interface between the RAN nodes **111** and MMEs.

(44) When a user equipment (UE) is operating in Evolved-Universal Terrestrial Radio Access-New Radio (EN-DC), a gNodeB (gNB) serving the UE configures the UE to send measurement reports for channel state information reference signal (CSI-RS) resources, Synchronization Signal Block (SSB) resources (also referred to as SS/Physical Broadcast Channel (PBCH) or SS/PBCH), or both CSI-RS and SSB resources. Each SSB resource may be associated with a beam from a plurality of beams called SSB beams. The UE searches for and measure the beams, maintaining a set of candidate beams. The candidate set of beams may contain beams from multiple cells of the network. Physical Cell ID (PCI) and beam ID are the identifications used to identify the beams.

(45) Based on the measurement reports, the gNB selects gNB transmitter (Tx) beams with which to serve the UE. Currently, these layer 1 reference signal receiving power (L1-RSRP) measurement based reports are calculated according to 3GPP 38.214. 3GPP 38.214 specifies aspects of the L1-RSRP measurement outlined below.

(46) 5.2.1.4.3 L1-RSRP Reporting

(47) For L1-RSRP computation, in some implementations, the UE may be configured with CSI-RS resources, SS/PBCH Block resources or both CSI-RS and SS/PBCH block resources, when resource-wise quasi co-located with ‘QCL-Type C’ and ‘QCL-TypeD’ when applicable.

(48) Additionally, in some implementations, the UE may be configured with CSI-RS resource setting up to 16 CSI-RS resource sets having up to 64 resources within each set. In some implementations, the total number of different CSI-RS resources over all resource sets is no more than 128. For L1-RSRP reporting, in some implementations, if the higher layer parameter `nrofReportedRS` in `CSI-ReportConfig` is configured to be one, the reported L1-RSRP value is defined by a 7-bit value in the range $[-140, -44]$ dBm with 1 dB step size, if the higher layer parameter `nrofReportedRS` is configured to be larger than one, or if the higher layer parameter `groupBasedBeamReporting` is configured as 'enabled', the UE shall use differential L1-RSRP based reporting, where the largest measured value of L1-RSRP is quantized to a 7-bit value in the range $[-140, -44]$ dBm with 1 dB step size, and the differential L1-RSRP is quantized to a 4-bit value. In some implementations, the differential L1-RSRP value is computed with 2 dB step size with a reference to the largest measured L1-RSRP value which is part of the same L1-RSRP reporting instance. In some implementations, the mapping between the reported L1-RSRP value and the measured quantity is described in [11, TS 38.133].

(49) Under this process, the UE measures RSRP for both horizontal and vertical polarization. Then, the UE calculates a filtered value that is reported to the gNB. The filtered value can be a maximum or of the two measurements or an addition of the two.

(50) In field and lab tests, it is found that there could be significant difference in measurements for horizontal and vertical polarization, which is referred to as an H/V imbalance (measured in decibels [dB]). It could be as high as 20-30 dB.

(51) Thus, using the current implementation of sub-optimal SSB reporting described above, the UE could end up on a sub-optimal Tx beam, which in turn, leads to poor downlink performance and NR link retention.

(52) FIG. 2 illustrates an example of data obtained via a lab test indicating that a UE switched to a beam having high H/V polarization imbalance, leading to poor DL performance and link failure. FIG. 3 illustrates an example of data obtained via a field test for pedestrian mobility indicating that a UE switched to a beam having high H/V polarization imbalance, leading to poor DL performance and link failure. These examples show that a UE that reported RSRP for a Tx beam for which it measured high imbalance, switched to that Tx beam as per gNB response to reported measurements. This eventually lead to poor DL performance and link failure.

(53) FIG. 4 is a flowchart of an example of a process 400 for enhanced SSB Beam RSRP reporting according to one aspect of the present disclosure. For clarity of presentation, the description that follows generally describes method 400 in the context of the other figures in this description. However, it will be understood that method 400 can be performed, for example, by any suitable system, environment, software, hardware, or a combination of systems, environments, software, and hardware, as appropriate. In some implementations, various steps of method 400 can be run in parallel, in combination, in loops, or in any order.

(54) At step 402, the UE determines that the UE is in EN-DC connected state with RSRP measurements configured. At step 404, the method involves the UE determining whether a highest SSB RSRP measurement is greater than a predetermined value, x (e.g., in dBm). If the highest SSB RSRP measurement is greater than the predetermined value, the UE moves to step 406. At step 406, the UE performs the default rank/reporting method (e.g., as specified in 3GPP 38.214). However, if the highest SSB RSRP measurement is less than or equal to the predetermined value, the UE moves to step 408. At step 408, the UE sorts and ranks the SSB measurements with respect to RSRP. At step 410, the UE sorts the RSRP measurements based on Rx beam type (e.g., in case it is different from step 408). At step 412, the UE calculates weight based rank for the SSBs as per RSRP and H/V imbalance. At step 414, the UE reports the SSB measurement that has the least H/V imbalance to the base station.

(55) This optimized SSB measurement reporting leads to switching to a better gNB Tx beam, which in turn, sustains NR performance and extends NR coverage.

(56) FIG. 5 is a flowchart of another example of a process 500 for enhanced SSB Beam RSRP reporting according to one aspect of the present disclosure. For clarity of presentation, the description that follows generally describes process 500 in the context of the other figures in this description. However, it will be understood that process 500 can be performed, for example, by any suitable system, environment, software, hardware, or a combination of systems, environments, software, and hardware, as appropriate. In some implementations, various steps of process 500 can be run in parallel, in combination, in loops, or in any order.

(57) A UE can begin execution of the process 500 by measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams from a base station (510).

(58) The UE can continue execution of the process 500 by determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold (520). If the UE determines that the greatest RSRP value is greater than the predetermined threshold at stage 520, the UE can determine to use default beam ranking/reporting methods. Default beam ranking/reporting methods may be any conventional or non-conventional methods for which the UE may be configured to perform beam ranking/reporting.

(59) Alternatively, responsive to the determination at stage 520 that the RSRP value is less than or equal to the predetermined threshold, the UE can continue execution of the process 500 by calculating, based at least in part on the measured RSRP values and a horizontal/vertical (H/V) polarization imbalance for each of the plurality of SSB beams, a ranking of the SSB plurality of beams (530). In some implementations, the calculating at stage 530 can include determining, for each particular SSB beam of the plurality of SSB beams, a weight based on the RSRP value for the particular SSB beam and the H/V polarization imbalance for the SSB beam, and calculating the ranking based on the determined weights for the plurality SSB beams.

(60) In some implementations, determining the horizontal/vertical (H/V) polarization imbalance for each particular SSB beam can include determining an horizontal RSRP value for a horizontal polarization of the particular SSB beam, determining a vertical RSRP value for vertical polarization of the particular SSB beam, and determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value.

(61) In some implementations, determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value can include (a) selecting a maximum of the horizontal RSRP value and the vertical RSRP value. In other implementations, determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value can include (b) selecting a sum of the horizontal RSRP value and the vertical RSRP value.

(62) The UE can continue execution of the process 500 by reporting at least one of the ranked plurality of SSB beams to the base station (540). In some implementations, reporting at least one of the ranked plurality of beams to the base station can include reporting a top ranked beam to the base station. In some implementations, the top ranked beam has a lowest H/V imbalance amongst the plurality of beams.

(63) In some implementations, the UE performing execution of the process 500 is operating in Evolved-Universal Terrestrial Radio Access-New Radio (EN-DC). In some implementations, the base station is one of a gNodeB (gNB) or an eNB.

(64) The example of process 500 described herein is described with respect to reference signals comprising SSB reference signals. However, the present disclosure is not so limited. Instead, the process 500 can be used to measure other reference signals and perform the remaining operations of process 500 with respect to other reference signals or values related thereto. For example, in some implementations, the process 500 can be used to measure RSRP values of channel state information reference signals (CSI-RS) and perform the remaining operations of process 500 on

CSI-RS reference signals (or values related thereto) instead of SSB reference signals (or values related thereto). In yet other implementations, the process **500** can be used on a combination of references signals (or values related thereto). For example, in some implementations, the process **500** can be performed on both SSB and CSI-RS reference signals.

(65) FIG. **6** illustrates an example architecture of a system **600** including a first CN **620**, in accordance with various embodiments. In this example, system **600** may implement the LTE standard wherein the CN **620** is an EPC **620** that corresponds with CN **120** of FIG. **1**. Additionally, the UE **601** may be the same or similar as the UEs **101** of FIG. **1**, and the E-UTRAN **610** may be a RAN that is the same or similar to the RAN **110** of FIG. **1**, and which may include RAN nodes **111** discussed previously. The CN **620** may comprise MMEs **621**, an S-GW **622**, a P-GW **623**, a HSS **624**, and a SGSN **625**.

(66) The MMEs **621** may be similar in function to the control plane of legacy SGSN, and may implement MM functions to keep track of the current location of a UE **601**. The MMEs **621** may perform various MM procedures to manage mobility aspects in access such as gateway selection and tracking area list management. MM (also referred to as “EPS MM” or “EMM” in E-UTRAN systems) may refer to all applicable procedures, methods, data storage, etc. That are used to maintain knowledge about a present location of the UE **601**, provide user identity confidentiality, and/or perform other like services to users/subscribers. Each UE **601** and the MME **621** may include an MM or EMM sublayer, and an MM context may be established in the UE **601** and the MME **621** when an attach procedure is successfully completed. The MM context may be a data structure or database object that stores MM-related information of the UE **601**. The MMEs **621** may be coupled with the HSS **624** via an S6a reference point, coupled with the SGSN **625** via an S3 reference point, and coupled with the S-GW **622** via an S11 reference point.

(67) The SGSN **625** may be a node that serves the UE **601** by tracking the location of an individual UE **601** and performing security functions. In addition, the SGSN **625** may perform Inter-EPC node signaling for mobility between 2G/3G and E-UTRAN 3GPP access networks; PDN and S-GW selection as specified by the MMEs **621**; handling of UE **601** time zone functions as specified by the MMEs **621**; and MME selection for handovers to E-UTRAN 3GPP access network. The S3 reference point between the MMEs **621** and the SGSN **625** may enable user and bearer information exchange for inter-3GPP access network mobility in idle and/or active states.

(68) The HSS **624** may comprise a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The EPC **620** may comprise one or several HSSs **624**, depending on the number of mobile subscribers, on the capacity of the equipment, on the organization of the network, etc. For example, the HSS **624** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **624** and the MMEs **621** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the EPC **620** between HSS **624** and the MMEs **621**.

(69) The S-GW **622** may terminate the S1 interface **113** (“S1-U” in FIG. **6**) toward the RAN **610**, and routes data packets between the RAN **610** and the EPC **620**. In addition, the S-GW **622** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement. The S11 reference point between the S-GW **622** and the MMEs **621** may provide a control plane between the MMEs **621** and the S-GW **622**. The S-GW **622** may be coupled with the P-GW **623** via an S5 reference point.

(70) The P-GW **623** may terminate an SGi interface toward a PDN **630**. The P-GW **623** may route data packets between the EPC **620** and external networks such as a network including the application server **130** (alternatively referred to as an “AF”) via an IP interface **125** (see e.g., FIG. **1**). In embodiments, the P-GW **623** may be communicatively coupled to an application server (application server **130** of FIG. **1** or PDN **630** in FIG. **6**) via an IP communications interface **125**

(see, e.g., FIG. 1). The S5 reference point between the P-GW **623** and the S-GW **622** may provide user plane tunneling and tunnel management between the P-GW **623** and the S-GW **622**. The S5 reference point may also be used for S-GW **622** relocation due to UE **601** mobility and if the S-GW **622** needs to connect to a non-collocated P-GW **623** for the required PDN connectivity. The P-GW **623** may further include a node for policy enforcement and charging data collection (e.g., PCEF (not shown)). Additionally, the SGi reference point between the P-GW **623** and the packet data network (PDN) **630** may be an operator external public, a private PDN, or an intra operator packet data network, for example, for provision of IMS services. The P-GW **623** may be coupled with a PCRF **626** via a Gx reference point.

(71) PCRF **626** is the policy and charging control element of the EPC **620**. In a non-roaming scenario, there may be a single PCRF **626** in the Home Public Land Mobile Network (HPLMN) associated with a UE **601**'s Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with local breakout of traffic, there may be two PCRFs associated with a UE **601**'s IP-CAN session, a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **626** may be communicatively coupled to the application server **630** via the P-GW **623**. The application server **630** may signal the PCRF **626** to indicate a new service flow and select the appropriate QoS and charging parameters. The PCRF **626** may provision this rule into a PCEF (not shown) with the appropriate TFT and QCI, which commences the QoS and charging as specified by the application server **630**. The Gx reference point between the PCRF **626** and the P-GW **623** may allow for the transfer of QoS policy and charging rules from the PCRF **626** to PCEF in the P-GW **623**. An Rx reference point may reside between the PDN **630** (or "AF **630**") and the PCRF **626**.

(72) FIG. 7 illustrates an architecture of a system **700** including a second CN **720** in accordance with various embodiments. The system **700** is shown to include a UE **701**, which may be the same or similar to the UEs **101** and UE **601** discussed previously; a (R)AN **710**, which may be the same or similar to the RAN **110** and RAN **610** discussed previously, and which may include RAN nodes **111** discussed previously; and a DN **703**, which may be, for example, operator services, Internet access or 3rd party services; and a 5GC **720**. The 5GC **720** may include an AUSF **722**; an AMF **721**; a SMF **724**; a NEF **723**; a PCF **726**; a NRF **725**; a UDM **727**; an AF **728**; a UPF **702**; and a NSSF **729**.

(73) The UPF **702** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to DN **703**, and a branching point to support multi-homed PDU session. The UPF **702** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform Uplink Traffic verification (e.g., SDF to QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **702** may include an uplink classifier to support routing traffic flows to a data network. The DN **703** may represent various network operator services, Internet access, or third party services. DN **703** may include, or be similar to, application server **130** discussed previously. The UPF **702** may interact with the SMF **724** via an N4 reference point between the SMF **724** and the UPF **702**.

(74) The AUSF **722** may store data for authentication of UE **701** and handle authentication-related functionality. The AUSF **722** may facilitate a common authentication framework for various access types. The AUSF **722** may communicate with the AMF **721** via an N12 reference point between the AMF **721** and the AUSF **722**; and may communicate with the UDM **727** via an N13 reference point between the UDM **727** and the AUSF **722**. Additionally, the AUSF **722** may exhibit an Nausf service-based interface.

(75) The AMF **721** may be responsible for registration management (e.g., for registering UE **701**, etc.), connection management, reachability management, mobility management, and lawful

interception of AMF-related events, and access authentication and authorization. The AMF 721 may be a termination point for the N11 reference point between the AMF 721 and the SMF 724. The AMF 721 may provide transport for SM messages between the UE 701 and the SMF 724, and act as a transparent proxy for routing SM messages. AMF 721 may also provide transport for SMS messages between UE 701 and an SMSF (not shown by FIG. 7). AMF 721 may act as SEAF, which may include interaction with the AUSF 722 and the UE 701, receipt of an intermediate key that was established as a result of the UE 701 authentication process. Where USIM based authentication is used, the AMF 721 may retrieve the security material from the AUSF 722. AMF 721 may also include a SCM function, which receives a key from the SEA that it uses to derive access-network specific keys. Furthermore, AMF 721 may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the (R)AN 710 and the AMF 721; and the AMF 721 may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection.

(76) AMF 721 may also support NAS signaling with a UE 701 over an N3 IWF interface. The N3IWF may be used to provide access to untrusted entities. N3IWF may be a termination point for the N2 interface between the (R)AN 710 and the AMF 721 for the control plane, and may be a termination point for the N3 reference point between the (R)AN 710 and the UPF 702 for the user plane. As such, the AMF 721 may handle N2 signaling from the SMF 724 and the AMF 721 for PDU sessions and QoS, encapsulate/de-encapsulate packets for IPsec and N3 tunneling, mark N3 user-plane packets in the uplink, and enforce QoS corresponding to N3 packet marking taking into account QoS requirements associated with such marking received over N2. N3 IWF may also relay uplink and downlink control-plane NAS signaling between the UE 701 and AMF 721 via an N1 reference point between the UE 701 and the AMF 721, and relay uplink and downlink user-plane packets between the UE 701 and UPF 702. The N3 IWF also provides mechanisms for IPsec tunnel establishment with the UE 701. The AMF 721 may exhibit an Namf service-based interface, and may be a termination point for an N14 reference point between two AMFs 721 and an N17 reference point between the AMF 721 and a 5G-EIR (not shown by FIG. 7).

(77) The UE 701 may need to register with the AMF 721 in order to receive network services. RM is used to register or deregister the UE 701 with the network (e.g., AMF 721), and establish a UE context in the network (e.g., AMF 721). The UE 701 may operate in an RM-REGISTERED state or an RM-DEREGISTERED state. In the RM DEREGISTERED state, the UE 701 is not registered with the network, and the UE context in AMF 721 holds no valid location or routing information for the UE 701 so the UE 701 is not reachable by the AMF 721. In the RM REGISTERED state, the UE 701 is registered with the network, and the UE context in AMF 721 may hold a valid location or routing information for the UE 701 so the UE 701 is reachable by the AMF 721. In the RM-REGISTERED state, the UE 701 may perform mobility Registration Update procedures, perform periodic Registration Update procedures triggered by expiration of the periodic update timer (e.g., to notify the network that the UE 701 is still active), and perform a Registration Update procedure to update UE capability information or to re-negotiate protocol parameters with the network, among others.

(78) The AMF 721 may store one or more RM contexts for the UE 701, where each RM context is associated with a specific access to the network. The RM context may be a data structure, database object, etc. That indicates or stores, inter alia, a registration state per access type and the periodic update timer. The AMF 721 may also store a 5GC MM context that may be the same or similar to the (E)MM context discussed previously. In various embodiments, the AMF 721 may store a CE mode B Restriction parameter of the UE 701 in an associated MM context or RM context. The AMF 721 may also derive the value, when needed, from the UE's usage setting parameter already stored in the UE context (and/or MM/RM context).

(79) CM may be used to establish and release a signaling connection between the UE 701 and the AMF 721 over the N1 interface. The signaling connection is used to enable NAS signaling

exchange between the UE **701** and the CN **720**, and comprises both the signaling connection between the UE and the AN (e.g., RRC connection or UE-N3 IWF connection for non-3GPP access) and the N2 connection for the UE **701** between the AN (e.g., RAN **710**) and the AMF **721**. The UE **701** may operate in one of two CM states, CM-IDLE mode or CM-CONNECTED mode. When the UE **701** is operating in the CM-IDLE state/mode, the UE **701** may have no NAS signaling connection established with the AMF **721** over the N1 interface, and there may be (R)AN **710** signaling connection (e.g., N2 and/or N3 connections) for the UE **701**. When the UE **701** is operating in the CM-CONNECTED state/mode, the UE **701** may have an established NAS signaling connection with the AMF **721** over the N1 interface, and there may be a (R)AN **710** signaling connection (e.g., N2 and/or N3 connections) for the UE **701**. Establishment of an N2 connection between the (R)AN **710** and the AMF **721** may cause the UE **701** to transition from CM-IDLE mode to CM-CONNECTED mode, and the UE **701** may transition from the CM-CONNECTED mode to the CM-IDLE mode when N2 signaling between the (R)AN **710** and the AMF **721** is released.

(80) The SMF **724** may be responsible for SM (e.g., session establishment, modify and release, including tunnel maintain between UPF and AN node); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF over N2 to AN; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or “session” may refer to a PDU connectivity service that provides or enables the exchange of PDUs between a UE **701** and a data network (DN) **703** identified by a Data Network Name (DNN). PDU sessions may be established upon UE **701** request, modified upon UE **701** and 5GC **720** request, and released upon UE **701** and 5GC **720** request using NAS SM signaling exchanged over the N1 reference point between the UE **701** and the SMF **724**. Upon request from an application server, the 5GC **720** may trigger a specific application in the UE **701**. In response to receipt of the trigger message, the UE **701** may pass the trigger message (or relevant parts/information of the trigger message) to one or more identified applications in the UE **701**. The identified application(s) in the UE **701** may establish a PDU session to a specific DNN. The SMF **724** may check whether the UE **701** requests are compliant with user subscription information associated with the UE **701**. In this regard, the SMF **724** may retrieve and/or request to receive update notifications on SMF **724** level subscription data from the UDM **727**.

(81) The SMF **724** may include the following roaming functionality: handling local enforcement to apply QoS SLAB (VPLMN); charging data collection and charging interface (VPLMN); lawful intercept (in VPLMN for SM events and interface to LI system); and support for interaction with external DN for transport of signaling for PDU session authorization/authentication by external DN. An N16 reference point between two SMF s **724** may be included in the system **700**, which may be between another SMF **724** in a visited network and the SMF **724** in the home network in roaming scenarios. Additionally, the SMF **724** may exhibit the Nsmf service-based interface.

(82) The NEF **723** may provide means for securely exposing the services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, Application Functions (e.g., AF **728**), edge computing or fog computing systems, etc. In such embodiments, the NEF **723** may authenticate, authorize, and/or throttle the AFs. NEF **723** may also translate information exchanged with the AF **728** and information exchanged with internal network functions. For example, the NEF **723** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **723** may also receive information from other network functions (NFs) based on exposed capabilities of other network functions. This information may be stored at the NEF **723** as structured data, or at a data storage NF using standardized interfaces. The stored information can

then be re-exposed by the NEF 723 to other NFs and AFs, and/or used for other purposes such as analytics. Additionally, the NEF 723 may exhibit an Nnef service-based interface.

(83) The NRF 725 may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF 725 also maintains information of available NF instances and their supported services. As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF 725 may exhibit the Nnrf service-based interface.

(84) The PCF 726 may provide policy rules to control plane function(s) to enforce them, and may also support unified policy framework to govern network behavior. The PCF 726 may also implement an FE to access subscription information relevant for policy decisions in a UDR of the UDM 727. The PCF 726 may communicate with the AMF 721 via an N15 reference point between the PCF 726 and the AMF 721, which may include a PCF 726 in a visited network and the AMF 721 in case of roaming scenarios. The PCF 726 may communicate with the AF 728 via an N5 reference point between the PCF 726 and the AF 728; and with the SMF 724 via an N7 reference point between the PCF 726 and the SMF 724. The system 700 and/or CN 720 may also include an N24 reference point between the PCF 726 (in the home network) and a PCF 726 in a visited network. Additionally, the PCF 726 may exhibit an Npcf service-based interface.

(85) The UDM 727 may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE 701. For example, subscription data may be communicated between the UDM 727 and the AMF 721 via an N8 reference point between the UDM 727 and the AMF. The UDM 727 may include two parts, an application FE and a UDR (the FE and UDR are not shown by FIG. 7). The UDR may store subscription data and policy data for the UDM 727 and the PCF 726, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs 701) for the NEF 723. The Nudr service-based interface may be exhibited by the UDR 221 to allow the UDM 727, PCF 726, and NEF 723 to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management, and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. The UDR may interact with the SMF 724 via an N10 reference point between the UDM 727 and the SMF 724. UDM 727 may also support SMS management, wherein an SMS-FE implements the similar application logic as discussed previously. Additionally, the UDM 727 may exhibit the Nudm service-based interface.

(86) The AF 728 may provide application influence on traffic routing, provide access to the NCE, and interact with the policy framework for policy control. The NCE may be a mechanism that allows the 5GC 720 and AF 728 to provide information to each other via NEF 723, which may be used for edge computing implementations. In such implementations, the network operator and third party services may be hosted close to the UE 701 access point of attachment to achieve an efficient service delivery through the reduced end-to-end latency and load on the transport network. For edge computing implementations, the 5GC may select a UPF 702 close to the UE 701 and execute traffic steering from the UPF 702 to DN 703 via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF 728. In this way, the AF 728 may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF 728 is considered to be a trusted entity, the network operator may permit AF 728 to interact directly with relevant NFs. Additionally, the AF 728 may exhibit an Naf service-based interface.

(87) The NSSF **729** may select a set of network slice instances serving the UE **701**. The NSSF **729** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **729** may also determine the AMF set to be used to serve the UE **701**, or a list of candidate AMF(s) **721** based on a suitable configuration and possibly by querying the NRF **725**. The selection of a set of network slice instances for the UE **701** may be triggered by the AMF **721** with which the UE **701** is registered by interacting with the NSSF **729**, which may lead to a change of AMF **721**. The NSSF **729** may interact with the AMF **721** via an N22 reference point between AMF **721** and NSSF **729**; and may communicate with another NSSF **729** in a visited network via an N31 reference point (not shown by FIG. 7). Additionally, the NSSF **729** may exhibit an Nnssf service-based interface.

(88) As discussed previously, the CN **720** may include an SMSF, which may be responsible for SMS subscription checking and verification, and relaying SM messages to/from the UE **701** to/from other entities, such as an SMS-GMSC/IW MSC/SMS-router. The SMS may also interact with AMF **721** and UDM **727** for a notification procedure that the UE **701** is available for SMS transfer (e.g., set a UE not reachable flag, and notifying UDM **727** when UE **701** is available for SMS).

(89) The CN **120** may also include other elements that are not shown by FIG. 7, such as a Data Storage system/architecture, a 5G-EIR, a SEPP, and the like. The Data Storage system may include a SDSF, an UDSF, and/or the like. Any NF may store and retrieve unstructured data into/from the UDSF (e.g., UE contexts), via N18 reference point between any NF and the UDSF (not shown by FIG. 7). Individual NFs may share a UDSF for storing their respective unstructured data or individual NFs may each have their own UDSF located at or near the individual NFs. Additionally, the UDSF may exhibit an Nudsf service-based interface (not shown by FIG. 7). The 5G-EIR may be an NF that checks the status of PEI for determining whether particular equipment/entities are blacklisted from the network; and the SEPP may be a non-transparent proxy that performs topology hiding, message filtering, and policing on inter-PLMN control plane interfaces.

(90) Additionally, there may be many more reference points and/or service-based interfaces between the NF services in the NFs; however, these interfaces and reference points have been omitted from FIG. 7 for clarity. In one example, the CN **720** may include an Nx interface, which is an inter-CN interface between the MME (e.g., MME **621**) and the AMF **721** in order to enable interworking between CN **720** and CN **620**. Other example interfaces/reference points may include an N5g-EIR service-based interface exhibited by a 5G-EIR, an N27 reference point between the NRF in the visited network and the NRF in the home network; and an N31 reference point between the NSSF in the visited network and the NSSF in the home network.

(91) FIG. 8 illustrates an example of infrastructure equipment **800** in accordance with various embodiments. The infrastructure equipment **800** (or “system **800**”) may be implemented as a base station, radio head, RAN node such as the RAN nodes **111** and/or AP **106** shown and described previously, application server(s) **130**, and/or any other element/device discussed herein. In other examples, the system **800** could be implemented in or by a UE.

(92) The system **800** includes application circuitry **805**, baseband circuitry **810**, one or more radio front end modules (RFEMs) **815**, memory circuitry **820**, power management integrated circuitry (PMIC) **825**, power tee circuitry **830**, network controller circuitry **835**, network interface connector **840**, satellite positioning circuitry **845**, and user interface **850**. In some embodiments, the device **800** may include additional elements such as, for example, memory/storage, display, camera, sensor, or input/output (I/O) interface. In other embodiments, the components described below may be included in more than one device. For example, said circuitries may be separately included in more than one device for CRAN, vBBU, or other like implementations.

(93) Application circuitry **805** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface

module, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose input/output (I/O or IO), memory card controllers such as Secure Digital (SD) MultiMediaCard (MMC) or similar, Universal Serial Bus (USB) interfaces, Mobile Industry Processor Interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors (or cores) of the application circuitry **805** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **800**. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

(94) The processor(s) of application circuitry **805** may include, for example, one or more processor cores (CPUs), one or more application processors, one or more graphics processing units (GPUs), one or more reduced instruction set computing (RISC) processors, one or more Acorn RISC Machine (ARM) processors, one or more complex instruction set computing (CISC) processors, one or more digital signal processors (DSP), one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, or any suitable combination thereof. In some embodiments, the application circuitry **805** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein. As examples, the processor(s) of application circuitry **805** may include one or more may include one or more Apple A-series processors, Intel Pentium®, Core®, or Xeon® processor(s); Advanced Micro Devices (AMD) Ryzen® processor(s), Accelerated Processing Units (APUs), or Epyc® processors; ARM-based processor(s) licensed from ARM Holdings, Ltd. Such as the ARM Cortex-A family of processors and the ThunderX2® provided by Cavium™, Inc.; a MIPS-based design from MIPS Technologies, Inc. Such as MIPS Warrior P-class processors; and/or the like. In some embodiments, the system **800** may not utilize application circuitry **805**, and instead may include a special-purpose processor/controller to process IP data received from an EPC or 5GC, for example.

(95) In some implementations, the application circuitry **805** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. As examples, the programmable processing devices may be one or more a field-programmable devices (FPDs) such as field-programmable gate arrays (FPGAs) and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such implementations, the circuitry of application circuitry **805** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. Of the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **805** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), anti-fuses, etc.)) used to store logic blocks, logic fabric, data, etc. In look-up-tables (LUTs) and the like.

(96) The baseband circuitry **810** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **810** are discussed infra with regard to FIG. **10**.

(97) User interface circuitry **850** may include one or more user interfaces designed to enable user interaction with the system **800** or peripheral component interfaces designed to enable peripheral component interaction with the system **800**. User interfaces may include, but are not limited to, one or more physical or virtual buttons (e.g., a reset button), one or more indicators (e.g., light emitting

diodes (LEDs)), a physical keyboard or keypad, a mouse, a touchpad, a touchscreen, speakers or other audio emitting devices, microphones, a printer, a scanner, a headset, a display screen or display device, etc. Peripheral component interfaces may include, but are not limited to, a nonvolatile memory port, a universal serial bus (USB) port, an audio jack, a power supply interface, etc.

(98) The radio front end modules (RFEMs) **815** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1011** of FIG. **10** infra), and the RFEM may be connected to multiple antennas. In alternative implementations, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **815**, which incorporates both mmWave antennas and sub-mmWave.

(99) The memory circuitry **820** may include one or more of volatile memory including dynamic random access memory (DRAM) and/or synchronous dynamic random access memory (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc., and may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®. Memory circuitry **820** may be implemented as one or more of solder down packaged integrated circuits, socketed memory modules and plug-in memory cards.

(100) The PMIC **825** may include voltage regulators, surge protectors, power alarm detection circuitry, and one or more backup power sources such as a battery or capacitor. The power alarm detection circuitry may detect one or more of brown out (under-voltage) and surge (over-voltage) conditions. The power tee circuitry **830** may provide for electrical power drawn from a network cable to provide both power supply and data connectivity to the infrastructure equipment **800** using a single cable.

(101) The network controller circuitry **835** may provide connectivity to a network using a standard network interface protocol such as Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), or some other suitable protocol. Network connectivity may be provided to/from the infrastructure equipment **800** via network interface connector **840** using a physical connection, which may be electrical (commonly referred to as a “copper interconnect”), optical, or wireless. The network controller circuitry **835** may include one or more dedicated processors and/or FPGAs to communicate using one or more of the aforementioned protocols. In some implementations, the network controller circuitry **835** may include multiple controllers to provide connectivity to other networks using the same or different protocols.

(102) The positioning circuitry **845** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a global navigation satellite system (GNSS). Examples of navigation satellite constellations (or GNSS) include United States' Global Positioning System (GPS), Russia's Global Navigation System (GLONASS), the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., Navigation with Indian Constellation (NAVIC), Japan's Quasi-Zenith Satellite System (QZSS), France's Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS), etc.), or the like. The positioning circuitry **845** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **845** may include a Micro-Technology for Positioning, Navigation, and Timing (Micro-PNT) IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **845** may also be part of, or interact with, the baseband

circuitry **810** and/or RFEMs **815** to communicate with the nodes and components of the positioning network. The positioning circuitry **845** may also provide position data and/or time data to the application circuitry **805**, which may use the data to synchronize operations with various infrastructure (e.g., RAN nodes **111**, etc.), or the like.

(103) The components shown by FIG. **8** may communicate with one another using interface circuitry, which may include any number of bus and/or interconnect (IX) technologies such as industry standard architecture (ISA), extended ISA (EISA), peripheral component interconnect (PCI), peripheral component interconnect extended (PCIx), PCI express (PCIe), or any number of other technologies. The bus/IX may be a proprietary bus, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I2C interface, an SPI interface, point to point interfaces, and a power bus, among others.

(104) FIG. **9** illustrates an example of a platform **900** (or “device **900**”) in accordance with various embodiments. In embodiments, the computer platform **900** may be suitable for use as UEs **101**, **601**, **701**, application servers **130**, and/or any other element/device discussed herein. The platform **900** may include any combinations of the components shown in the example. The components of platform **900** may be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules, logic, hardware, software, firmware, or a combination thereof adapted in the computer platform **900**, or as components otherwise incorporated within a chassis of a larger system. The block diagram of FIG. **9** is intended to show a high level view of components of the computer platform **900**. However, some of the components shown may be omitted, additional components may be present, and different arrangement of the components shown may occur in other implementations.

(105) Application circuitry **905** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of LDOs, interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, RTC, timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as SD MMC or similar, USB interfaces, MIPI interfaces, and JTAG test access ports. The processors (or cores) of the application circuitry **905** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **900**. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

(106) The processor(s) of application circuitry **805** may include, for example, one or more processor cores, one or more application processors, one or more GPUs, one or more RISC processors, one or more ARM processors, one or more CISC processors, one or more DSP, one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, a multithreaded processor, an ultra-low voltage processor, an embedded processor, some other known processing element, or any suitable combination thereof. In some embodiments, the application circuitry **805** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein.

(107) As examples, the processor(s) of application circuitry **905** may include an Apple A-series processor. The processors of the application circuitry **905** may also be one or more of an Intel® Architecture Core™ based processor, such as a Quark™, an Atom™, an i3, an i5, an i7, or an MCU-class processor, or another such processor available from Intel® Corporation, Santa Clara, CA; Advanced Micro Devices (AMD) Ryzen® processor(s) or Accelerated Processing Units (APUs); Snapdragon™ processor(s) from Qualcomm® Technologies, Inc., Texas Instruments, Inc.® Open Multimedia Applications Platform (OMAP)™ processor(s); a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior M-class, Warrior I-class, and Warrior P-class

processors; an ARM-based design licensed from ARM Holdings, Ltd., such as the ARM Cortex-A, Cortex-R, and Cortex-M family of processors; or the like. In some implementations, the application circuitry **905** may be a part of a system on a chip (SoC) in which the application circuitry **905** and other components are formed into a single integrated circuit.

(108) Additionally or alternatively, application circuitry **905** may include circuitry such as, but not limited to, one or more a field-programmable devices (FPDs) such as FPGAs and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such embodiments, the circuitry of application circuitry **905** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. Of the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **905** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), anti-fuses, etc.)) used to store logic blocks, logic fabric, data, etc. In look-up tables (LUTs) and the like.

(109) The baseband circuitry **910** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **910** are discussed infra with regard to FIG. **10**.

(110) The RFEMs **915** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1011** of FIG. **10** infra), and the RFEM may be connected to multiple antennas. In alternative implementations, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **915**, which incorporates both mmWave antennas and sub-mmWave.

(111) The memory circuitry **920** may include any number and type of memory devices used to provide for a given amount of system memory. As examples, the memory circuitry **920** may include one or more of volatile memory including random access memory (RAM), dynamic RAM (DRAM) and/or synchronous dynamic RAM (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc. The memory circuitry **920** may be developed in accordance with a Joint Electron Devices Engineering Council (JEDEC) low power double data rate (LPDDR)-based design, such as LPDDR2, LPDDR3, LPDDR4, or the like. Memory circuitry **920** may be implemented as one or more of solder down packaged integrated circuits, single die package (SDP), dual die package (DDP) or quad die package (Q17P), socketed memory modules, dual inline memory modules (DIMMs) including microDIMMs or MiniDIMMs, and/or soldered onto a motherboard via a ball grid array (BGA). In low power implementations, the memory circuitry **920** may be on-die memory or registers associated with the application circuitry **905**. To provide for persistent storage of information such as data, applications, operating systems and so forth, memory circuitry **920** may include one or more mass storage devices, which may include, inter alia, a solid state disk drive (SSDD), hard disk drive (HDD), a micro HDD, resistance change memories, phase change memories, holographic memories, or chemical memories, among others. For example, the computer platform **900** may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®.

(112) Removable memory circuitry **923** may include devices, circuitry, enclosures/housings, ports or receptacles, etc. Used to couple portable data storage devices with the platform **900**. These portable data storage devices may be used for mass storage purposes, and may include, for

example, flash memory cards (e.g., Secure Digital (SD) cards, microSD cards, xD picture cards, and the like), and USB flash drives, optical discs, external HDDs, and the like.

(113) The platform **900** may also include interface circuitry (not shown) that is used to connect external devices with the platform **900**. The external devices connected to the platform **900** via the interface circuitry include sensor circuitry **921** and electro-mechanical components (EMCs) **922**, as well as removable memory devices coupled to removable memory circuitry **923**.

(114) The sensor circuitry **921** include devices, modules, or subsystems whose purpose is to detect events or changes in its environment and send the information (sensor data) about the detected events to some other a device, module, subsystem, etc. Examples of such sensors include, inter alia, inertia measurement units (IMUs) comprising accelerometers, gyroscopes, and/or magnetometers; microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS) comprising 3-axis accelerometers, 3-axis gyroscopes, and/or magnetometers; level sensors; flow sensors; temperature sensors (e.g., thermistors); pressure sensors; barometric pressure sensors; gravimeters; altimeters; image capture devices (e.g., cameras or lensless apertures); light detection and ranging (LiDAR) sensors; proximity sensors (e.g., infrared radiation detector and the like), depth sensors, ambient light sensors, ultrasonic transceivers; microphones or other like audio capture devices; etc.

(115) EMCs **922** include devices, modules, or subsystems whose purpose is to enable platform **900** to change its state, position, and/or orientation, or move or control a mechanism or (sub)system. Additionally, EMCs **922** may be configured to generate and send messages/signaling to other components of the platform **900** to indicate a current state of the EMCs **922**. Examples of the EMCs **922** include one or more power switches, relays including electromechanical relays (EMRs) and/or solid state relays (SSRs), actuators (e.g., valve actuators, etc.), an audible sound generator, a visual warning device, motors (e.g., DC motors, stepper motors, etc.), wheels, thrusters, propellers, claws, clamps, hooks, and/or other like electro-mechanical components. In embodiments, platform **900** is configured to operate one or more EMCs **922** based on one or more captured events and/or instructions or control signals received from a service provider and/or various clients.

(116) In some implementations, the interface circuitry may connect the platform **900** with positioning circuitry **945**. The positioning circuitry **945** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a GNSS. Examples of navigation satellite constellations (or GNSS) include United States' GPS, Russia's GLONASS, the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., NAVIC), Japan's QZSS, France's DORIS, etc.), or the like. The positioning circuitry **945** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **945** may include a Micro-PNT IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **945** may also be part of, or interact with, the baseband circuitry **810** and/or RFEMs **915** to communicate with the nodes and components of the positioning network. The positioning circuitry **945** may also provide position data and/or time data to the application circuitry **905**, which may use the data to synchronize operations with various infrastructure (e.g., radio base stations), for turn-by-turn navigation applications, or the like

(117) In some implementations, the interface circuitry may connect the platform **900** with Near-Field Communication (NFC) circuitry **940**. NFC circuitry **940** is configured to provide contactless, short-range communications based on radio frequency identification (RFID) standards, wherein magnetic field induction is used to enable communication between NFC circuitry **940** and NFC-enabled devices external to the platform **900** (e.g., an “NFC touchpoint”). NFC circuitry **940** comprises an NFC controller coupled with an antenna element and a processor coupled with the NFC controller. The NFC controller may be a chip/IC providing NFC functionalities to the NFC circuitry **940** by executing NFC controller firmware and an NFC stack. The NFC stack may be

executed by the processor to control the NFC controller, and the NFC controller firmware may be executed by the NFC controller to control the antenna element to emit short-range RF signals. The RF signals may power a passive NFC tag (e.g., a microchip embedded in a sticker or wristband) to transmit stored data to the NFC circuitry **940**, or initiate data transfer between the NFC circuitry **940** and another active NFC device (e.g., a smartphone or an NFC-enabled POS terminal) that is proximate to the platform **900**.

(118) The driver circuitry **946** may include software and hardware elements that operate to control particular devices that are embedded in the platform **900**, attached to the platform **900**, or otherwise communicatively coupled with the platform **900**. The driver circuitry **946** may include individual drivers allowing other components of the platform **900** to interact with or control various input/output (I/O) devices that may be present within, or connected to, the platform **900**. For example, driver circuitry **946** may include a display driver to control and allow access to a display device, a touchscreen driver to control and allow access to a touchscreen interface of the platform **900**, sensor drivers to obtain sensor readings of sensor circuitry **921** and control and allow access to sensor circuitry **921**, EMC drivers to obtain actuator positions of the EMCs **922** and/or control and allow access to the EMCs **922**, a camera driver to control and allow access to an embedded image capture device, audio drivers to control and allow access to one or more audio devices.

(119) The power management integrated circuitry (PMIC) **925** (also referred to as “power management circuitry **925**”) may manage power provided to various components of the platform **900**. In particular, with respect to the baseband circuitry **910**, the PMIC **925** may control power-source selection, voltage scaling, battery charging, or DC-to-DC conversion. The PMIC **925** may often be included when the platform **900** is capable of being powered by a battery **930**, for example, when the device is included in a UE **101**, **601**, **701**.

(120) In some embodiments, the PMIC **925** may control, or otherwise be part of, various power saving mechanisms of the platform **900**. For example, if the platform **900** is in an RRC_Connected state, where it is still connected to the RAN node as it expects to receive traffic shortly, then it may enter a state known as Discontinuous Reception Mode (DRX) after a period of inactivity. During this state, the platform **900** may power down for brief intervals of time and thus save power. If there is no data traffic activity for an extended period of time, then the platform **900** may transition off to an RRC Idle state, where it disconnects from the network and does not perform operations such as channel quality feedback, handover, etc. The platform **900** goes into a very low power state and it performs paging where again it periodically wakes up to listen to the network and then powers down again. The platform **900** may not receive data in this state; in order to receive data, it must transition back to RRC_Connected state. An additional power saving mode may allow a device to be unavailable to the network for periods longer than a paging interval (ranging from seconds to a few hours). During this time, the device is totally unreachable to the network and may power down completely. Any data sent during this time incurs a large delay and it is assumed the delay is acceptable.

(121) A battery **930** may power the platform **900**, although in some examples the platform **900** may be mounted deployed in a fixed location, and may have a power supply coupled to an electrical grid. The battery **930** may be a lithium ion battery, a metal-air battery, such as a zinc-air battery, an aluminum-air battery, a lithium-air battery, and the like. In some implementations, such as in V2X applications, the battery **930** may be a typical lead-acid automotive battery.

(122) In some implementations, the battery **930** may be a “smart battery,” which includes or is coupled with a Battery Management System (BMS) or battery monitoring integrated circuitry. The BMS may be included in the platform **900** to track the state of charge (SoCh) of the battery **930**. The BMS may be used to monitor other parameters of the battery **930** to provide failure predictions, such as the state of health (SoH) and the state of function (SoF) of the battery **930**. The BMS may communicate the information of the battery **930** to the application circuitry **905** or other components of the platform **900**. The BMS may also include an analog-to-digital (ADC) convertor

that allows the application circuitry **905** to directly monitor the voltage of the battery **930** or the current flow from the battery **930**. The battery parameters may be used to determine actions that the platform **900** may perform, such as transmission frequency, network operation, sensing frequency, and the like.

(123) A power block, or other power supply coupled to an electrical grid may be coupled with the BMS to charge the battery **930**. In some examples, the power block **XS30** may be replaced with a wireless power receiver to obtain the power wirelessly, for example, through a loop antenna in the computer platform **900**. In these examples, a wireless battery charging circuit may be included in the BMS. The specific charging circuits chosen may depend on the size of the battery **930**, and thus, the current required. The charging may be performed using the Airfuel standard promulgated by the Airfuel Alliance, the Qi wireless charging standard promulgated by the Wireless Power Consortium, or the Rezence charging standard promulgated by the Alliance for Wireless Power, among others.

(124) User interface circuitry **950** includes various input/output (I/O) devices present within, or connected to, the platform **900**, and includes one or more user interfaces designed to enable user interaction with the platform **900** and/or peripheral component interfaces designed to enable peripheral component interaction with the platform **900**. The user interface circuitry **950** includes input device circuitry and output device circuitry. Input device circuitry includes any physical or virtual means for accepting an input including, inter alia, one or more physical or virtual buttons (e.g., a reset button), a physical keyboard, keypad, mouse, touchpad, touchscreen, microphones, scanner, headset, and/or the like. The output device circuitry includes any physical or virtual means for showing information or otherwise conveying information, such as sensor readings, actuator position(s), or other like information. Output device circuitry may include any number and/or combinations of audio or visual display, including, inter alia, one or more simple visual outputs/indicators (e.g., binary status indicators (e.g., light emitting diodes (LEDs)) and multi-character visual outputs, or more complex outputs such as display devices or touchscreens (e.g., Liquid Chrystal Displays (LCD), LED displays, quantum dot displays, projectors, etc.), with the output of characters, graphics, multimedia objects, and the like being generated or produced from the operation of the platform **900**. The output device circuitry may also include speakers or other audio emitting devices, printer(s), and/or the like. In some embodiments, the sensor circuitry **921** may be used as the input device circuitry (e.g., an image capture device, motion capture device, or the like) and one or more EMCs may be used as the output device circuitry (e.g., an actuator to provide haptic feedback or the like). In another example, NFC circuitry comprising an NFC controller coupled with an antenna element and a processing device may be included to read electronic tags and/or connect with another NFC-enabled device. Peripheral component interfaces may include, but are not limited to, a non-volatile memory port, a USB port, an audio jack, a power supply interface, etc.

(125) Although not shown, the components of platform **900** may communicate with one another using a suitable bus or interconnect (IX) technology, which may include any number of technologies, including ISA, EISA, PCI, PCIX, PCIe, a Time-Trigger Protocol (TTP) system, a FlexRay system, or any number of other technologies. The bus/IX may be a proprietary bus/IX, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I2C interface, an SPI interface, point-to-point interfaces, and a power bus, among others.

(126) FIG. **10** illustrates example components of baseband circuitry **1010** and radio front end modules (RFEM) **1015** in accordance with various embodiments. The baseband circuitry **1010** corresponds to the baseband circuitry **810** and **910** of FIGS. **8** and **9**, respectively. The RFEM **1015** corresponds to the RFEM **815** and **915** of FIGS. **8** and **9**, respectively. As shown, the RFEMs **1015** may include Radio Frequency (RF) circuitry **1006**, front-end module (FEM) circuitry **1008**, antenna array **1011** coupled together at least as shown.

(127) The baseband circuitry **1010** includes circuitry and/or control logic configured to carry out

various radio/network protocol and radio control functions that enable communication with one or more radio networks via the RF circuitry **1006**. The radio control functions may include, but are not limited to, signal modulation/demodulation, encoding/decoding, radio frequency shifting, etc. In some embodiments, modulation/demodulation circuitry of the baseband circuitry **1010** may include Fast-Fourier Transform (FFT), precoding, or constellation mapping/demapping functionality. In some embodiments, encoding/decoding circuitry of the baseband circuitry **1010** may include convolution, tail-biting convolution, turbo, Viterbi, or Low Density Parity Check (LDPC) encoder/decoder functionality. Embodiments of modulation/demodulation and encoder/decoder functionality are not limited to these examples and may include other suitable functionality in other embodiments. The baseband circuitry **1010** is configured to process baseband signals received from a receive signal path of the RF circuitry **1006** and to generate baseband signals for a transmit signal path of the RF circuitry **1006**. The baseband circuitry **1010** is configured to interface with application circuitry **805/XS205** (see FIGS. **8** and **9**) for generation and processing of the baseband signals and for controlling operations of the RF circuitry **1006**. The baseband circuitry **1010** may handle various radio control functions.

(128) The aforementioned circuitry and/or control logic of the baseband circuitry **1010** may include one or more single or multi-core processors. For example, the one or more processors may include a 3G baseband processor **1004A**, a 4G/LTE baseband processor **1004B**, a 5G/NR baseband processor **1004C**, or some other baseband processor(s) **1004D** for other existing generations, generations in development or to be developed in the future (e.g., sixth generation (6G), etc.). In other embodiments, some or all of the functionality of baseband processors **1004A-D** may be included in modules stored in the memory **1004G** and executed via a Central Processing Unit (CPU) **1004E**. In other embodiments, some or all of the functionality of baseband processors **1004A-D** may be provided as hardware accelerators (e.g., FPGAs, ASICs, etc.) loaded with the appropriate bit streams or logic blocks stored in respective memory cells. In various embodiments, the memory **1004G** may store program code of a real-time OS (RTOS), which when executed by the CPU **1004E** (or other baseband processor), is to cause the CPU **1004E** (or other baseband processor) to manage resources of the baseband circuitry **1010**, schedule tasks, etc. Examples of the RTOS may include Operating System Embedded (OSE)[™] provided by Enea®, Nucleus RTOS[™] provided by Mentor Graphics®, Versatile Real-Time Executive (VRTX) provided by Mentor Graphics®, ThreadX[™] provided by Express Logic®, FreeRTOS, REX OS provided by Qualcomm®, OKL4 provided by Open Kernel (OK) Labs®, or any other suitable RTOS, such as those discussed herein. In addition, the baseband circuitry **1010** includes one or more audio digital signal processor(s) (DSP) **1004F**. The audio DSP(s) **1004F** include elements for compression/decompression and echo cancellation and may include other suitable processing elements in other embodiments.

(129) In some embodiments, each of the processors **1004A-XT104E** include respective memory interfaces to send/receive data to/from the memory **1004G**. The baseband circuitry **1010** may further include one or more interfaces to communicatively couple to other circuitries/devices, such as an interface to send/receive data to/from memory external to the baseband circuitry **1010**; an application circuitry interface to send/receive data to/from the application circuitry **805/XS205** of FIGS. **8-XT**); an RF circuitry interface to send/receive data to/from RF circuitry **1006** of FIG. **10**; a wireless hardware connectivity interface to send/receive data to/from one or more wireless hardware elements (e.g., Near Field Communication (NFC) components, Bluetooth®/Bluetooth® Low Energy components, Wi-Fi® components, and/or the like); and a power management interface to send/receive power or control signals to/from the PMIC **925**.

(130) In alternate embodiments (which may be combined with the above described embodiments), baseband circuitry **1010** comprises one or more digital baseband systems, which are coupled with one another via an interconnect subsystem and to a CPU subsystem, an audio subsystem, and an interface subsystem. The digital baseband subsystems may also be coupled to a digital baseband

interface and a mixed-signal baseband subsystem via another interconnect subsystem. Each of the interconnect subsystems may include a bus system, point-to-point connections, network-on-chip (NOC) structures, and/or some other suitable bus or interconnect technology, such as those discussed herein. The audio subsystem may include DSP circuitry, buffer memory, program memory, speech processing accelerator circuitry, data converter circuitry such as analog-to-digital and digital-to-analog converter circuitry, analog circuitry including one or more of amplifiers and filters, and/or other like components. In an aspect of the present disclosure, baseband circuitry **1010** may include protocol processing circuitry with one or more instances of control circuitry (not shown) to provide control functions for the digital baseband circuitry and/or radio frequency circuitry (e.g., the radio front end modules **1015**).

(131) Although not shown by FIG. **10**, in some embodiments, the baseband circuitry **1010** includes individual processing device(s) to operate one or more wireless communication protocols (e.g., a “multi-protocol baseband processor” or “protocol processing circuitry”) and individual processing device(s) to implement PHY layer functions. In these embodiments, the PHY layer functions include the aforementioned radio control functions. In these embodiments, the protocol processing circuitry operates or implements various protocol layers/entities of one or more wireless communication protocols. In a first example, the protocol processing circuitry may operate LTE protocol entities and/or 5G/NR protocol entities when the baseband circuitry **1010** and/or RF circuitry **1006** are part of mmWave communication circuitry or some other suitable cellular communication circuitry. In the first example, the protocol processing circuitry would operate MAC, RLC, PDCP, SDAP, RRC, and NAS functions. In a second example, the protocol processing circuitry may operate one or more IEEE-based protocols when the baseband circuitry **1010** and/or RF circuitry **1006** are part of a Wi-Fi communication system. In the second example, the protocol processing circuitry would operate Wi-Fi MAC and logical link control (LLC) functions. The protocol processing circuitry may include one or more memory structures (e.g., **1004G**) to store program code and data for operating the protocol functions, as well as one or more processing cores to execute the program code and perform various operations using the data. The baseband circuitry **1010** may also support radio communications for more than one wireless protocol.

(132) The various hardware elements of the baseband circuitry **1010** discussed herein may be implemented, for example, as a solder-down substrate including one or more integrated circuits (ICs), a single packaged IC soldered to a main circuit board or a multi-chip module containing two or more ICs. In one example, the components of the baseband circuitry **1010** may be suitably combined in a single chip or chipset, or disposed on a same circuit board. In another example, some or all of the constituent components of the baseband circuitry **1010** and RF circuitry **1006** may be implemented together such as, for example, a system on a chip (SoC) or System-in-Package (SiP). In another example, some or all of the constituent components of the baseband circuitry **1010** may be implemented as a separate SoC that is communicatively coupled with and RF circuitry **1006** (or multiple instances of RF circuitry **1006**). In yet another example, some or all of the constituent components of the baseband circuitry **1010** and the application circuitry **805/XS205** may be implemented together as individual SoCs mounted to a same circuit board (e.g., a “multi-chip package”).

(133) In some embodiments, the baseband circuitry **1010** may provide for communication compatible with one or more radio technologies. For example, in some embodiments, the baseband circuitry **1010** may support communication with an E-UTRAN or other WMAN, a WLAN, a WPAN. Embodiments in which the baseband circuitry **1010** is configured to support radio communications of more than one wireless protocol may be referred to as multi-mode baseband circuitry.

(134) RF circuitry **1006** may enable communication with wireless networks using modulated electromagnetic radiation through a non-solid medium. In various embodiments, the RF circuitry **1006** may include switches, filters, amplifiers, etc. To facilitate the communication with the

wireless network. RF circuitry **1006** may include a receive signal path, which may include circuitry to down-convert RF signals received from the FEM circuitry **1008** and provide baseband signals to the baseband circuitry **1010**. RF circuitry **1006** may also include a transmit signal path, which may include circuitry to up-convert baseband signals provided by the baseband circuitry **1010** and provide RF output signals to the FEM circuitry **1008** for transmission.

(135) In some embodiments, the receive signal path of the RF circuitry **1006** may include mixer circuitry **1006a**, amplifier circuitry **1006b** and filter circuitry **1006c**. In some embodiments, the transmit signal path of the RF circuitry **1006** may include filter circuitry **1006c** and mixer circuitry **1006a**. RF circuitry **1006** may also include synthesizer circuitry **1006d** for synthesizing a frequency for use by the mixer circuitry **1006a** of the receive signal path and the transmit signal path. In some embodiments, the mixer circuitry **1006a** of the receive signal path may be configured to down-convert RF signals received from the FEM circuitry **1008** based on the synthesized frequency provided by synthesizer circuitry **1006d**. The amplifier circuitry **1006b** may be configured to amplify the down-converted signals and the filter circuitry **1006c** may be a low-pass filter (LPF) or band-pass filter (BPF) configured to remove unwanted signals from the down-converted signals to generate output baseband signals. Output baseband signals may be provided to the baseband circuitry **1010** for further processing. In some embodiments, the output baseband signals may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **1006a** of the receive signal path may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

(136) In some embodiments, the mixer circuitry **1006a** of the transmit signal path may be configured to up-convert input baseband signals based on the synthesized frequency provided by the synthesizer circuitry **1006d** to generate RF output signals for the FEM circuitry **1008**. The baseband signals may be provided by the baseband circuitry **1010** and may be filtered by filter circuitry **1006c**.

(137) In some embodiments, the mixer circuitry **1006a** of the receive signal path and the mixer circuitry **1006a** of the transmit signal path may include two or more mixers and may be arranged for quadrature downconversion and upconversion, respectively. In some embodiments, the mixer circuitry **1006a** of the receive signal path and the mixer circuitry **1006a** of the transmit signal path may include two or more mixers and may be arranged for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1006a** of the receive signal path and the mixer circuitry **1006a** of the transmit signal path may be arranged for direct downconversion and direct upconversion, respectively. In some embodiments, the mixer circuitry **1006a** of the receive signal path and the mixer circuitry **1006a** of the transmit signal path may be configured for super-heterodyne operation.

(138) In some embodiments, the output baseband signals and the input baseband signals may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals and the input baseband signals may be digital baseband signals. In these alternate embodiments, the RF circuitry **1006** may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry and the baseband circuitry **1010** may include a digital baseband interface to communicate with the RF circuitry **1006**.

(139) In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, although the scope of the embodiments is not limited in this respect.

(140) In some embodiments, the synthesizer circuitry **1006d** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1006d** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider.

(141) The synthesizer circuitry **1006d** may be configured to synthesize an output frequency for use by the mixer circuitry **1006a** of the RF circuitry **1006** based on a frequency input and a divider control input. In some embodiments, the synthesizer circuitry **1006d** may be a fractional $N/N+1$ synthesizer.

(142) In some embodiments, frequency input may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. Divider control input may be provided by either the baseband circuitry **1010** or the application circuitry **805/XS205** depending on the desired output frequency. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table based on a channel indicated by the application circuitry **805/XS205**.

(143) Synthesizer circuitry **1006d** of the RF circuitry **1006** may include a divider, a delay-locked loop (DLL), a multiplexer and a phase accumulator. In some embodiments, the divider may be a dual modulus divider (DMD) and the phase accumulator may be a digital phase accumulator (DPA). In some embodiments, the DMD may be configured to divide the input signal by either N or $N+1$ (e.g., based on a carry out) to provide a fractional division ratio. In some example embodiments, the DLL may include a set of cascaded, tunable, delay elements, a phase detector, a charge pump and a D-type flip-flop. In these embodiments, the delay elements may be configured to break a VCO period up into N_d equal packets of phase, where N_d is the number of delay elements in the delay line. In this way, the DLL provides negative feedback to help ensure that the total delay through the delay line is one VCO cycle.

(144) In some embodiments, synthesizer circuitry **1006d** may be configured to generate a carrier frequency as the output frequency, while in other embodiments, the output frequency may be a multiple of the carrier frequency (e.g., twice the carrier frequency, four times the carrier frequency) and used in conjunction with quadrature generator and divider circuitry to generate multiple signals at the carrier frequency with multiple different phases with respect to each other. In some embodiments, the output frequency may be a LO frequency (fLO). In some embodiments, the RF circuitry **1006** may include an IQ/polar converter.

(145) FEM circuitry **1008** may include a receive signal path, which may include circuitry configured to operate on RF signals received from antenna array **1011**, amplify the received signals and provide the amplified versions of the received signals to the RF circuitry **1006** for further processing. FEM circuitry **1008** may also include a transmit signal path, which may include circuitry configured to amplify signals for transmission provided by the RF circuitry **1006** for transmission by one or more of antenna elements of antenna array **1011**. In various embodiments, the amplification through the transmit or receive signal paths may be done solely in the RF circuitry **1006**, solely in the FEM circuitry **1008**, or in both the RF circuitry **1006** and the FEM circuitry **1008**.

(146) In some embodiments, the FEM circuitry **1008** may include a TX/RX switch to switch between transmit mode and receive mode operation. The FEM circuitry **1008** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **1008** may include an LNA to amplify received RF signals and provide the amplified received RF signals as an output (e.g., to the RF circuitry **1006**). The transmit signal path of the FEM circuitry **1008** may include a power amplifier (PA) to amplify input RF signals (e.g., provided by RF circuitry **1006**), and one or more filters to generate RF signals for subsequent transmission by one or more antenna elements of the antenna array **1011**.

(147) The antenna array **1011** comprises one or more antenna elements, each of which is configured convert electrical signals into radio waves to travel through the air and to convert received radio waves into electrical signals. For example, digital baseband signals provided by the baseband circuitry **1010** is converted into analog RF signals (e.g., modulated waveform) that will be amplified and transmitted via the antenna elements of the antenna array **1011** including one or more antenna elements (not shown). The antenna elements may be omnidirectional, direction, or a combination thereof. The antenna elements may be formed in a multitude of arrangements as are known

and/or discussed herein. The antenna array **1011** may comprise microstrip antennas or printed antennas that are fabricated on the surface of one or more printed circuit boards. The antenna array **1011** may be formed in as a patch of metal foil (e.g., a patch antenna) in a variety of shapes, and may be coupled with the RF circuitry **1006** and/or FEM circuitry **1008** using metal transmission lines or the like.

(148) Processors of the application circuitry **805/XS205** and processors of the baseband circuitry **1010** may be used to execute elements of one or more instances of a protocol stack. For example, processors of the baseband circuitry **1010**, alone or in combination, may be used execute Layer 3, Layer 2, or Layer 1 functionality, while processors of the application circuitry **805/XS205** may utilize data (e.g., packet data) received from these layers and further execute Layer 4 functionality (e.g., TCP and UDP layers). As referred to herein, Layer 3 may comprise a RRC layer, described in further detail below. As referred to herein, Layer 2 may comprise a MAC layer, an RLC layer, and a PDCP layer, described in further detail below. As referred to herein, Layer 1 may comprise a PHY layer of a UE/RAN node, described in further detail below.

(149) FIG. **11** illustrates various protocol functions that may be implemented in a wireless communication device according to various embodiments. In particular, FIG. **11** includes an arrangement **1100** showing interconnections between various protocol layers/entities. The following description of FIG. **11** is provided for various protocol layers/entities that operate in conjunction with the 5G/NR system standards and LTE system standards, but some or all of the aspects of FIG. **11** may be applicable to other wireless communication network systems as well.

(150) The protocol layers of arrangement **1100** may include one or more of PHY **1110**, MAC **1120**, RLC **1130**, PDCP **1140**, SDAP **1147**, RRC **1155**, and NAS layer **1157**, in addition to other higher layer functions not illustrated. The protocol layers may include one or more service access points (e.g., items **1159**, **1156**, **1150**, **1149**, **1145**, **1135**, **1125**, and **1115** in FIG. **11**) that may provide communication between two or more protocol layers.

(151) The PHY **1110** may transmit and receive physical layer signals **1105** that may be received from or transmitted to one or more other communication devices. The physical layer signals **1105** may comprise one or more physical channels, such as those discussed herein. The PHY **1110** may further perform link adaptation or adaptive modulation and coding (AMC), power control, cell search (e.g., for initial synchronization and handover purposes), and other measurements used by higher layers, such as the RRC **1155**. The PHY **1110** may still further perform error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, modulation/demodulation of physical channels, interleaving, rate matching, mapping onto physical channels, and MIMO antenna processing. In embodiments, an instance of PHY **1110** may process requests from and provide indications to an instance of MAC **1120** via one or more PHY-SAP **1115**. According to some embodiments, requests and indications communicated via PHY-SAP **1115** may comprise one or more transport channels.

(152) Instance(s) of MAC **1120** may process requests from, and provide indications to, an instance of RLC **1130** via one or more MAC-SAPs **1125**. These requests and indications communicated via the MAC-SAP **1125** may comprise one or more logical channels. The MAC **1120** may perform mapping between the logical channels and transport channels, multiplexing of MAC SDUs from one or more logical channels onto TBs to be delivered to PHY **1110** via the transport channels, demultiplexing MAC SDUs to one or more logical channels from TBs delivered from the PHY **1110** via transport channels, multiplexing MAC SDUs onto TBs, scheduling information reporting, error correction through HARQ, and logical channel prioritization.

(153) Instance(s) of RLC **1130** may process requests from and provide indications to an instance of PDCP **1140** via one or more radio link control service access points (RLC-SAP) **1135**. These requests and indications communicated via RLC-SAP **1135** may comprise one or more RLC channels. The RLC **1130** may operate in a plurality of modes of operation, including: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). The RLC **1130** may

execute transfer of upper layer protocol data units (PDUs), error correction through automatic repeat request (ARQ) for AM data transfers, and concatenation, segmentation and reassembly of RLC SDUs for UM and AM data transfers. The RLC **1130** may also execute re-segmentation of RLC data PDUs for AM data transfers, reorder RLC data PDUs for UM and AM data transfers, detect duplicate data for UM and AM data transfers, discard RLC SDUs for UM and AM data transfers, detect protocol errors for AM data transfers, and perform RLC re-establishment.

(154) Instance(s) of PDCP **1140** may process requests from and provide indications to instance(s) of RRC **1155** and/or instance(s) of SDAP **1147** via one or more packet data convergence protocol service access points (PDCP-SAP) **1145**. These requests and indications communicated via PDCP-SAP **1145** may comprise one or more radio bearers. The PDCP **1140** may execute header compression and decompression of IP data, maintain PDCP Sequence Numbers (SNs), perform in-sequence delivery of upper layer PDUs at re-establishment of lower layers, eliminate duplicates of lower layer SDUs at re-establishment of lower layers for radio bearers mapped on RLC AM, cipher and decipher control plane data, perform integrity protection and integrity verification of control plane data, control timer-based discard of data, and perform security operations (e.g., ciphering, deciphering, integrity protection, integrity verification, etc.).

(155) Instance(s) of SDAP **1147** may process requests from and provide indications to one or more higher layer protocol entities via one or more SDAP-SAP **1149**. These requests and indications communicated via SDAP-SAP **1149** may comprise one or more QoS flows. The SDAP **1147** may map QoS flows to DRBs, and vice versa, and may also mark QFIs in DL and UL packets. A single SDAP entity **1147** may be configured for an individual PDU session. In the UL direction, the NG-RAN **110** may control the mapping of QoS Flows to DRB(s) in two different ways, reflective mapping or explicit mapping. For reflective mapping, the SDAP **1147** of a UE **101** may monitor the QFIs of the DL packets for each DRB, and may apply the same mapping for packets flowing in the UL direction. For a DRB, the SDAP **1147** of the UE **101** may map the UL packets belonging to the QoS flows(s) corresponding to the QoS flow ID(s) and PDU session observed in the DL packets for that DRB. To enable reflective mapping, the NG-RAN **710** may mark DL packets over the Uu interface with a QoS flow ID. The explicit mapping may involve the RRC **1155** configuring the SDAP **1147** with an explicit QoS flow to DRB mapping rule, which may be stored and followed by the SDAP **1147**. In embodiments, the SDAP **1147** may only be used in NR implementations and may not be used in LTE implementations.

(156) The RRC **1155** may configure, via one or more management service access points (M-SAP), aspects of one or more protocol layers, which may include one or more instances of PHY **1110**, MAC **1120**, RLC **1130**, PDCP **1140** and SDAP **1147**. In embodiments, an instance of RRC **1155** may process requests from and provide indications to one or more NAS entities **1157** via one or more RRC-SAPs **1156**. The main services and functions of the RRC **1155** may include broadcast of system information (e.g., included in MIBs or SIBs related to the NAS), broadcast of system information related to the access stratum (AS), paging, establishment, maintenance and release of an RRC connection between the UE **101** and RAN **110** (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), establishment, configuration, maintenance and release of point to point Radio Bearers, security functions including key management, inter-RAT mobility, and measurement configuration for UE measurement reporting. The MIBs and SIBs may comprise one or more IEs, which may each comprise individual data fields or data structures.

(157) The NAS **1157** may form the highest stratum of the control plane between the UE **101** and the AMF **721**. The NAS **1157** may support the mobility of the UEs **101** and the session management procedures to establish and maintain IP connectivity between the UE **101** and a P-GW in LTE systems.

(158) According to various embodiments, one or more protocol entities of arrangement **1100** may be implemented in UEs **101**, RAN nodes **111**, AMF **721** in NR implementations or MME **621** in

LTE implementations, UPF **702** in NR implementations or S-GW **622** and P-GW **623** in LTE implementations, or the like to be used for control plane or user plane communications protocol stack between the aforementioned devices. In such embodiments, one or more protocol entities that may be implemented in one or more of UE **101**, gNB **111**, AMF **721**, etc. May communicate with a respective peer protocol entity that may be implemented in or on another device using the services of respective lower layer protocol entities to perform such communication. In some embodiments, a gNB-CU of the gNB **111** may host the RRC **1155**, SDAP **1147**, and PDCP **1140** of the gNB that controls the operation of one or more gNB-DUs, and the gNB-DUs of the gNB **111** may each host the RLC **1130**, MAC **1120**, and PHY **1110** of the gNB **111**.

(159) In a first example, a control plane protocol stack may comprise, in order from highest layer to lowest layer, NAS **1157**, RRC **1155**, PDCP **1140**, RLC **1130**, MAC **1120**, and PHY **1110**. In this example, upper layers **1160** may be built on top of the NAS **1157**, which includes an IP layer **1161**, an SCTP **1162**, and an application layer signaling protocol (AP) **1163**.

(160) In NR implementations, the AP **1163** may be an NG application protocol layer (NGAP or NG-AP) **1163** for the NG interface **113** defined between the NG-RAN node **111** and the AMF **721**, or the AP **1163** may be an Xn application protocol layer (XnAP or Xn-AP) **1163** for the Xn interface **112** that is defined between two or more RAN nodes **111**.

(161) The NG-AP **1163** may support the functions of the NG interface **113** and may comprise Elementary Procedures (EPs). An NG-AP EP may be a unit of interaction between the NG-RAN node **111** and the AMF **721**. The NG-AP **1163** services may comprise two groups: UE-associated services (e.g., services related to a UE **101**) and non-UE-associated services (e.g., services related to the whole NG interface instance between the NG-RAN node **111** and AMF **721**). These services may include functions including, but not limited to: a paging function for the sending of paging requests to NG-RAN nodes **111** involved in a particular paging area; a UE context management function for allowing the AMF **721** to establish, modify, and/or release a UE context in the AMF **721** and the NG-RAN node **111**; a mobility function for UEs **101** in ECM-CONNECTED mode for intra-system HOs to support mobility within NG-RAN and inter-system HOs to support mobility from/to EPS systems; a NAS Signaling Transport function for transporting or rerouting NAS messages between UE **101** and AMF **721**; a NAS node selection function for determining an association between the AMF **721** and the UE **101**; NG interface management function(s) for setting up the NG interface and monitoring for errors over the NG interface; a warning message transmission function for providing means to transfer warning messages via NG interface or cancel ongoing broadcast of warning messages; a Configuration Transfer function for requesting and transferring of RAN configuration information (e.g., SON information, performance measurement (PM) data, etc.) between two RAN nodes **111** via CN **120**; and/or other like functions.

(162) The XnAP **1163** may support the functions of the Xn interface **112** and may comprise XnAP basic mobility procedures and XnAP global procedures. The XnAP basic mobility procedures may comprise procedures used to handle UE mobility within the NG RAN **111** (or E-UTRAN **610**), such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The XnAP global procedures may comprise procedures that are not related to a specific UE **101**, such as Xn interface setup and reset procedures, NG-RAN update procedures, cell activation procedures, and the like.

(163) In LTE implementations, the AP **1163** may be an S1 Application Protocol layer (S1-AP) **1163** for the S1 interface **113** defined between an E-UTRAN node **111** and an MME, or the AP **1163** may be an X2 application protocol layer (X2AP or X2-AP) **1163** for the X2 interface **112** that is defined between two or more E-UTRAN nodes **111**.

(164) The S1 Application Protocol layer (S1-AP) **1163** may support the functions of the S1 interface, and similar to the NG-AP discussed previously, the S1-AP may comprise S1-AP EPs. An S1-AP EP may be a unit of interaction between the E-UTRAN node **111** and an MME **621** within

an LTE CN **120**. The S1-AP **1163** services may comprise two groups: UE-associated services and non UE-associated services. These services perform functions including, but not limited to: E-UTRAN Radio Access Bearer (E-RAB) management, UE capability indication, mobility, NAS signaling transport, RAN Information Management (RIM), and configuration transfer.

(165) The X2AP **1163** may support the functions of the X2 interface **112** and may comprise X2AP basic mobility procedures and X2AP global procedures. The X2AP basic mobility procedures may comprise procedures used to handle UE mobility within the E-UTRAN **120**, such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The X2AP global procedures may comprise procedures that are not related to a specific UE **101**, such as X2 interface setup and reset procedures, load indication procedures, error indication procedures, cell activation procedures, and the like.

(166) The SCTP layer (alternatively referred to as the SCTP/IP layer) **1162** may provide guaranteed delivery of application layer messages (e.g., NGAP or XnAP messages in NR implementations, or S1-AP or X2AP messages in LTE implementations). The SCTP **1162** may ensure reliable delivery of signaling messages between the RAN node **111** and the AMF **721**/MME **621** based, in part, on the IP protocol, supported by the IP **1161**. The Internet Protocol layer (IP) **1161** may be used to perform packet addressing and routing functionality. In some implementations the IP layer **1161** may use point-to-point transmission to deliver and convey PDUs. In this regard, the RAN node **111** may comprise L2 and L1 layer communication links (e.g., wired or wireless) with the MME/AMF to exchange information.

(167) In a second example, a user plane protocol stack may comprise, in order from highest layer to lowest layer, SDAP **1147**, PDCP **1140**, RLC **1130**, MAC **1120**, and PHY **1110**. The user plane protocol stack may be used for communication between the UE **101**, the RAN node **111**, and UPF **702** in NR implementations or an S-GW **622** and P-GW **623** in LTE implementations. In this example, upper layers **1151** may be built on top of the SDAP **1147**, and may include a user datagram protocol (UDP) and IP security layer (UDP/IP) **1152**, a General Packet Radio Service (GPRS) Tunneling Protocol for the user plane layer (GTP-U) **1153**, and a User Plane PDU layer (UP PDU) **1163**.

(168) The transport network layer **1154** (also referred to as a “transport layer”) may be built on IP transport, and the GTP-U **1153** may be used on top of the UDP/IP layer **1152** (comprising a UDP layer and IP layer) to carry user plane PDUs (UP-PDUs). The IP layer (also referred to as the “Internet layer”) may be used to perform packet addressing and routing functionality. The IP layer may assign IP addresses to user data packets in any of IPv4, IPv6, or PPP formats, for example.

(169) The GTP-U **1153** may be used for carrying user data within the GPRS core network and between the radio access network and the core network. The user data transported can be packets in any of IPv4, IPv6, or PPP formats, for example. The UDP/IP **1152** may provide checksums for data integrity, port numbers for addressing different functions at the source and destination, and encryption and authentication on the selected data flows. The RAN node **111** and the S-GW **622** may utilize an S1-U interface to exchange user plane data via a protocol stack comprising an L1 layer (e.g., PHY **1110**), an L2 layer (e.g., MAC **1120**, RLC **1130**, PDCP **1140**, and/or SDAP **1147**), the UDP/IP layer **1152**, and the GTP-U **1153**. The S-GW **622** and the P-GW **623** may utilize an S5/S8a interface to exchange user plane data via a protocol stack comprising an L1 layer, an L2 layer, the UDP/IP layer **1152**, and the GTP-U **1153**. As discussed previously, NAS protocols may support the mobility of the UE **101** and the session management procedures to establish and maintain IP connectivity between the UE **101** and the P-GW **623**.

(170) Moreover, although not shown by FIG. **11**, an application layer may be present above the AP **1163** and/or the transport network layer **1154**. The application layer may be a layer in which a user of the UE **101**, RAN node **111**, or other network element interacts with software applications being executed, for example, by application circuitry **805** or application circuitry **905**, respectively. The

application layer may also provide one or more interfaces for software applications to interact with communications systems of the UE **101** or RAN node **111**, such as the baseband circuitry **1010**. In some implementations the IP layer and/or the application layer may provide the same or similar functionality as layers 5-7, or portions thereof, of the Open Systems Interconnection (OSI) model (e.g., OSI Layer 7—the application layer, OSI Layer 6—the presentation layer, and OSI Layer 5—the session layer).

(171) FIG. **12** illustrates components of a core network in accordance with various embodiments. The components of the CN **620** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In embodiments, the components of CN **720** may be implemented in a same or similar manner as discussed herein with regard to the components of CN **620**. In some embodiments, NFV is utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **620** may be referred to as a network slice **1201**, and individual logical instantiations of the CN **620** may provide specific network capabilities and network characteristics. A logical instantiation of a portion of the CN **620** may be referred to as a network sub-slice **1202** (e.g., the network sub-slice **1202** is shown to include the P-GW **623** and the PCRF **626**).

(172) As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. A network instance may refer to information identifying a domain, which may be used for traffic detection and routing in case of different IP domains or overlapping IP addresses. A network slice instance may refer to a set of network functions (NFs) instances and the resources (e.g., compute, storage, and networking resources) required to deploy the network slice.

(173) With respect to 5G systems (see, e.g., FIG. **7**), a network slice always comprises a RAN part and a CN part. The support of network slicing relies on the principle that traffic for different slices is handled by different PDU sessions. The network can realize the different network slices by scheduling and also by providing different L1/L2 configurations. The UE **701** provides assistance information for network slice selection in an appropriate RRC message, if it has been provided by NAS. While the network can support large number of slices, the UE need not support more than 8 slices simultaneously.

(174) A network slice may include the CN **720** control plane and user plane NFs, NG-RANs **710** in a serving PLMN, and a N3 IWF functions in the serving PLMN. Individual network slices may have different S-NSSAI and/or may have different SSTs. NSSAI includes one or more S-NSSAIs, and each network slice is uniquely identified by an S-NSSAI. Network slices may differ for supported features and network functions optimizations, and/or multiple network slice instances may deliver the same service/features but for different groups of UEs **701** (e.g., enterprise users). For example, individual network slices may deliver different committed service(s) and/or may be dedicated to a particular customer or enterprise. In this example, each network slice may have different S-NSSAIs with the same SST but with different slice differentiators. Additionally, a single UE may be served with one or more network slice instances simultaneously via a 5G AN and associated with eight different S-NSSAIs. Moreover, an AMF **721** instance serving an individual UE **701** may belong to each of the network slice instances serving that UE.

(175) Network Slicing in the NG-RAN **710** involves RAN slice awareness. RAN slice awareness includes differentiated handling of traffic for different network slices, which have been pre-configured. Slice awareness in the NG-RAN **710** is introduced at the PDU session level by indicating the S-NSSAI corresponding to a PDU session in all signaling that includes PDU session resource information. How the NG-RAN **710** supports the slice enabling in terms of NG-RAN

functions (e.g., the set of network functions that comprise each slice) is implementation dependent. The NG-RAN **710** selects the RAN part of the network slice using assistance information provided by the UE **701** or the 5GC **720**, which unambiguously identifies one or more of the pre-configured network slices in the PLMN. The NG-RAN **710** also supports resource management and policy enforcement between slices as per SLAs. A single NG-RAN node may support multiple slices, and the NG-RAN **710** may also apply an appropriate RRM policy for the SLA in place to each supported slice. The NG-RAN **710** may also support QoS differentiation within a slice.

(176) The NG-RAN **710** may also use the UE assistance information for the selection of an AMF **721** during an initial attach, if available. The NG-RAN **710** uses the assistance information for routing the initial NAS to an AMF **721**. If the NG-RAN **710** is unable to select an AMF **721** using the assistance information, or the UE **701** does not provide any such information, the NG-RAN **710** sends the NAS signaling to a default AMF **721**, which may be among a pool of AMFs **721**. For subsequent accesses, the UE **701** provides a temp ID, which is assigned to the UE **701** by the 5GC **720**, to enable the NG-RAN **710** to route the NAS message to the appropriate AMF **721** as long as the temp ID is valid. The NG-RAN **710** is aware of, and can reach, the AMF **721** that is associated with the temp ID. Otherwise, the method for initial attach applies.

(177) The NG-RAN **710** supports resource isolation between slices. NG-RAN **710** resource isolation may be achieved by means of RRM policies and protection mechanisms that should avoid that shortage of shared resources if one slice breaks the service level agreement for another slice. In some implementations, it is possible to fully dedicate NG-RAN **710** resources to a certain slice. How NG-RAN **710** supports resource isolation is implementation dependent.

(178) Some slices may be available only in part of the network. Awareness in the NG-RAN **710** of the slices supported in the cells of its neighbors may be beneficial for inter-frequency mobility in connected mode. The slice availability may not change within the UE's registration area. The NG-RAN **710** and the 5GC **720** are responsible to handle a service request for a slice that may or may not be available in a given area. Admission or rejection of access to a slice may depend on factors such as support for the slice, availability of resources, support of the requested service by NG-RAN **710**.

(179) The UE **701** may be associated with multiple network slices simultaneously. In case the UE **701** is associated with multiple slices simultaneously, only one signaling connection is maintained, and for intra-frequency cell reselection, the UE **701** tries to camp on the best cell. For inter-frequency cell reselection, dedicated priorities can be used to control the frequency on which the UE **701** camps. The 5GC **720** is to validate that the UE **701** has the rights to access a network slice. Prior to receiving an Initial Context Setup Request message, the NG-RAN **710** may be allowed to apply some provisional/local policies, based on awareness of a particular slice that the UE **701** is requesting to access. During the initial context setup, the NG-RAN **710** is informed of the slice for which resources are being requested.

(180) NFV architectures and infrastructures may be used to virtualize one or more NFs, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

(181) FIG. **13** is a block diagram illustrating components, according to some example embodiments, of a system **1300** to support NFV. The system **1300** is illustrated as including a VIM **1302**, an NFVI **1304**, an VNFM **1306**, VNFs **1308**, an EM **1310**, an NFVO **1312**, and a NM **1314**.

(182) The VIM **1302** manages the resources of the NFVI **1304**. The NFVI **1304** can include physical or virtual resources and applications (including hypervisors) used to execute the system **1300**. The VIM **1302** may manage the life cycle of virtual resources with the NFVI **1304** (e.g., creation, maintenance, and tear down of VMs associated with one or more physical resources), track VM instances, track performance, fault and security of VM instances and associated physical

resources, and expose VM instances and associated physical resources to other management systems.

(183) The VNFM **1306** may manage the VNFs **1308**. The VNFs **1308** may be used to execute EPC components/functions. The VNFM **1306** may manage the life cycle of the VNFs **1308** and track performance, fault and security of the virtual aspects of VNFs **1308**. The EM **1310** may track the performance, fault and security of the functional aspects of VNFs **1308**. The tracking data from the VNFM **1306** and the EM **1310** may comprise, for example, PM data used by the VIM **1302** or the NFVI **1304**. Both the VNFM **1306** and the EM **1310** can scale up/down the quantity of VNFs of the system **1300**.

(184) The NFVO **1312** may coordinate, authorize, release and engage resources of the NFVI **1304** in order to provide the requested service (e.g., to execute an EPC function, component, or slice). The NM **1314** may provide a package of end-user functions with the responsibility for the management of a network, which may include network elements with VNFs, non-virtualized network functions, or both (management of the VNFs may occur via the EM **1310**).

(185) FIG. **14** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **14** shows a diagrammatic representation of hardware resources **1400** including one or more processors (or processor cores) **1410**, one or more memory/storage devices **1420**, and one or more communication resources **1430**, each of which may be communicatively coupled via a bus **1440**. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **1402** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **1400**.

(186) The processors **1410** may include, for example, a processor **1412** and a processor **1414**. The processor(s) **1410** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

(187) The memory/storage devices **1420** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **1420** may include, but are not limited to, any type of volatile or nonvolatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

(188) The communication resources **1430** may include interconnection or network interface components or other suitable devices to communicate with one or more peripheral devices **1404** or one or more databases **1406** via a network **1408**. For example, the communication resources **1430** may include wired communication components (e.g., for coupling via USB), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

(189) Instructions **1450** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **1410** to perform any one or more of the methodologies discussed herein. The instructions **1450** may reside, completely or partially, within at least one of the processors **1410** (e.g., within the processor's cache memory), the memory/storage devices **1420**, or any suitable combination thereof. Furthermore, any portion of the instructions **1450** may be transferred to the hardware resources **1400** from any combination of the peripheral devices **1404** or the databases **1406**. Accordingly, the memory of processors **1410**, the memory/storage devices **1420**, the peripheral devices **1404**, and the databases **1406** are examples of computer-readable and machine-readable media.

(190) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

EXAMPLES

(191) Example 1 may include a method for reference signal measurement reporting by a user equipment (UE) in a wireless communications system, the method involves: measuring, for each particular beam of a plurality of beams from a base station, a reference signal received power (RSRP) value for one or more reference signals associated with the particular beam, determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold, responsive to the determination, calculating, based at least in part on the measured RSRP values and a horizontal/vertical (H/V) polarization imbalance for each of the plurality of beams, a ranking of the plurality of beams, and reporting at least one of the ranked plurality of beams to the base station.

(192) Example 2 may include the method of example 1 and/or some other examples herein, where calculating, based at least in part on the measured RSRP values and a horizontal/vertical (H/V) polarization imbalance for each of the plurality of beams, a ranking of the plurality of beams involves determining, for each particular SSB beam of the plurality of SSB beams, a weight based on the RSRP value for the particular SSB beam and the H/V polarization imbalance for the SSB beam; and calculating the ranking based on the determined weights for the plurality SSB beams.

(193) Example 3 may include the method of examples 1-2 and/or some other examples herein, where reporting at least one of the ranked plurality of beams to the base station comprises reporting a top ranked beam to the base station.

(194) Example 4 may include the method of examples 1-3 and/or some other examples herein, where the top ranked beam has a lowest H/V imbalance amongst the plurality of beams.

(195) Example 5 may include the method of examples 1-4 and/or some other examples herein, where the one or more reference signals are at least one of channel state information reference signal (CSI-RS) resources or Synchronization Signal Block (SSB) resources.

(196) Example 6 may include the method of examples 1-5 and/or some other examples herein, where the UE is operating in Evolved-Universal Terrestrial Radio Access-New Radio (EN-DC).

(197) Example 7 may include the method of examples 1-6 and/or some other examples herein, where the base station is one of a gNodeB (gNB) or an eNB.

(198) Example 8 may include the method of claims 1-7 and/or some other examples herein, where determining the horizontal/vertical (H/V) polarization imbalance for each particular beam involves determining an horizontal RSRP value for a horizontal polarization of the particular beam; determining a vertical RSRP value for vertical polarization of the particular beam; and determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value.

(199) Example 9 may include the method of claims 1-8 and/or some other examples herein, where determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value involves selecting a maximum of the horizontal RSRP value and the vertical RSRP value.

(200) Example 10 may include the method of claims 1-9 and/or some other examples herein, where determining the horizontal/vertical (H/V) polarization imbalance for the particular beam based on the (i) the horizontal RSRP value and the (ii) the vertical RSRP value involves selecting a sum of the horizontal RSRP value and the vertical RSRP value.

(201) Example 11 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-10, or any other method or process described

herein.

(202) Example 12 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-10, or any other method or process described herein.

(203) Example 13 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-10, or any other method or process described herein.

(204) Example 14 may include a method, technique, or process as described in or related to any of examples 1-10, or portions or parts thereof.

(205) Example 15 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-10, or portions thereof.

(206) Example 16 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples 1-10, or portions thereof.

(207) Example 17 may include a system for providing wireless communication as shown and described herein.

(208) Example 18 may include a device for providing wireless communication as shown and described herein.

(209) Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Claims

1. A method comprising: measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams; in response to determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold: determining a horizontal RSRP value for a horizontal polarization of each particular SSB beam of the plurality of SSB beams; determining a vertical RSRP value for a vertical polarization of each particular SSB beam of the plurality of SSB beams; and determining a horizontal/vertical (H/V) polarization imbalance for each particular SSB beam of the plurality of SSB beams based at least on the horizontal RSRP value and the vertical RSRP value; and calculating, based at least on the determined H/V polarization imbalance, a ranking of the plurality of SSB beams; and reporting at least one of the ranked plurality of SSB beams.
2. The method of claim 1, wherein calculating the ranking of the plurality of SSB beams comprises: determining, for each particular SSB beam of the plurality of SSB beams, a weight based on the RSRP value for the particular SSB beam and the H/V polarization imbalance for the SSB beam; and calculating the ranking based on the determined weights for the plurality of SSB beams.
3. The method of claim 1, wherein reporting at least one of the ranked plurality of SSB beams comprises reporting a top ranked SSB beam of the plurality of SSB beams.
4. The method of claim 3, wherein the top ranked SSB beam has a lowest H/V imbalance amongst the plurality of SSB beams.
5. The method of claim 1, wherein the ranking is calculated by a user equipment (UE) that is operating in an Evolved-Universal Terrestrial Radio Access-New Radio (EN-DC) mode.
6. The method of claim 1, wherein the ranking is reported to a base station that is one of a gNodeB

(gNB) or an eNB.

7. The method of claim 1, wherein determining the H/V polarization imbalance comprises: (a) selecting a maximum of the horizontal RSRP value and the vertical RSRP value, or (b) selecting a sum of the horizontal RSRP value and the vertical RSRP value.

8. One or more processors configured to perform operations comprising: measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams; in response to determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold: determining a horizontal RSRP value for a horizontal polarization of each particular SSB beam of the plurality of SSB beams; determining a vertical RSRP value for a vertical polarization of each particular SSB beam of the plurality of SSB beams; and determining a horizontal/vertical (H/V) polarization imbalance for each particular SSB beam of the plurality of SSB beams based at least on the horizontal RSRP value and the vertical RSRP value; and calculating, based at least on the determined H/V polarization imbalance, a ranking of the plurality of SSB beams; and reporting at least one of the ranked plurality of SSB beams.

9. The one or more processors of claim 8, wherein calculating the ranking of the plurality of SSB beams comprises: determining, for each particular SSB beam of the plurality of SSB beams, a weight based on the RSRP value for the particular SSB beam and the H/V polarization imbalance for the SSB beam; and calculating the ranking based on the determined weights for the plurality of SSB beams.

10. The one or more processors of claim 8, wherein reporting at least one of the ranked plurality of SSB beams comprises reporting a top ranked SSB beam of the plurality of SSB beams.

11. The one or more processors of claim 10, wherein the top ranked SSB beam has a lowest H/V imbalance amongst the plurality of SSB beams.

12. The one or more processors of claim 8, wherein the ranking is calculated by a user equipment (UE) that is operating in an Evolved-Universal Terrestrial Radio Access-New Radio (EN-DC) mode.

13. The one or more processors of claim 8, wherein the ranking is reported to a base station that is one of a gNodeB (gNB) or an eNB.

14. The one or more processors of claim 8, wherein determining the H/V polarization imbalance comprises: (a) selecting a maximum of the horizontal RSRP value and the vertical RSRP value, or (b) selecting a sum of the horizontal RSRP value and the vertical RSRP value.

15. A non-transitory computer-readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: measuring a reference signal received power (RSRP) value for each of a plurality of synchronization signal block (SSB) beams; in response to determining that a greatest RSRP value of the measured RSRP values is less than or equal to a predetermined threshold: determining a horizontal RSRP value for a horizontal polarization of each particular SSB beam of the plurality of SSB beams; determining a vertical RSRP value for a vertical polarization of each particular SSB beam of the plurality of SSB beams; and determining a horizontal/vertical (H/V) polarization imbalance for each particular SSB beam of the plurality of SSB beams based at least on the horizontal RSRP value and the vertical RSRP value; and calculating, based at least on the determined H/V polarization imbalance, a ranking of the plurality of SSB beams; and reporting at least one of the ranked plurality of SSB beams.

16. The non-transitory computer-readable storage medium of claim 15, wherein calculating the ranking of the plurality of SSB beams comprises: determining, for each particular SSB beam of the plurality of SSB beams, a weight based on the RSRP value for the particular SSB beam and the H/V polarization imbalance for the SSB beam; and calculating the ranking based on the determined weights for the plurality of SSB beams.

17. The non-transitory computer-readable storage medium of claim 15, wherein determining the H/V polarization imbalance comprises: (a) selecting a maximum of the horizontal RSRP value and

the vertical RSRP value, or (b) selecting a sum of the horizontal RSRP value and the vertical RSRP value.
